

Default Options and Market Power: Evidence from Target-Date Funds*

Marco Loseto[†] Hanbin Yang[‡]

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Abstract

We study how default options generate market power in the U.S. retirement savings market. Focusing on target-date funds (TDFs), the predominant default in 401(k) plans, we show that TDFs charge substantially higher fees than their underlying funds, by 22 basis points or over 100% as of 2019. We argue that TDF managers exercise market power and price discriminate against fee-insensitive default investors. We then estimate a structural model of demand and fee-setting in TDF and non-TDF markets. Eliminating price discrimination between TDFs and non-TDFs increases TDF investors' welfare by \$1.2 billion per year with a modest reduction in managers' profits.

KEYWORDS: TDF, default options, 401(k) plans, fund fees.

JEL Classification: G11, G23, G28, G40, G51, J32, L51.

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[†]Bocconi University and IGIER, marco.loseto@unibocconi.it

[‡]London Business School, hyang@london.edu

1 Introduction

Default options in employer-sponsored 401(k) plans have transformed how American investors save for retirement. A large body of research shows that appropriate default funds, when combined with automatic enrollment, offset investors' inertia and inattention, leading to higher participation, savings, and equity exposure (Madrian and Shea, 2001; Choi et al., 2004; Beshears et al., 2009).

Since the 2006 Pension Protection Act (PPA), target-date funds (TDFs) have become the dominant default option in 401(k) plans, with nearly four trillion dollars in assets as of 2024.¹ Structured around an expected retirement year, TDFs gradually rebalance allocations from equities toward bonds as investors approach retirement, aligning portfolio risk with individuals' changing risk tolerance, human capital, and consumption needs over the life cycle (Viceira, 2001; Campbell and Viceira, 2002). While the benefits of TDFs as the default options are well established, much less is known about the frictions that shape competition and fee-setting of these products. Precisely because default investment options cater to investor inertia and inattention, they may also create scope for fund managers to exercise market power.

This paper studies how default investment options can create market power, allowing fund managers to extract rents from fee-insensitive investors. We show that fund managers charge substantially higher fees on TDFs than on the underlying funds held by those TDFs. Because TDFs are organized as funds-of-funds, managers can sell the same basket of funds both directly in the mutual fund market and indirectly through TDFs. Since many investors hold TDFs as their default investment and rarely rebalance, demand for TDFs tends to be relatively insensitive to fees. This creates scope for fund managers to price discriminate by charging higher fees on TDFs while maintaining lower fees on the underlying funds to compete for more fee-sensitive investors.

We begin by documenting three stylized facts about TDFs. First, we show that TDFs charge substantial fees on top of the fees of their underlying funds. We call this fee difference the *wrapper fee*, which is 22 bps on average as of 2019. There is also large variation in wrapper fees across funds, with those at the 90th percentile charging around 60 bps. Despite the secular decline in mutual fund fees, wrapper fees remain consistently over 100% of underlying fund fees and cost investors over \$3 billion each year.

Second, although wrapper fees could compensate fund managers for the added costs or value of managing TDFs, several pieces of evidence suggest that these factors account for only a small share of wrapper fees. TDFs provide life-cycle rebalancing, but this service relies on simple predetermined glide paths. As a comparison, TDF wrapper fees are substantially higher than index-fund expense ratios, even though index funds also follow rule-based rebalancing strategies, often across many more underlying securities.² In the

¹According to the Investment Company Institute, over 92% of TDF assets are held in retirement accounts.

²While TDF rebalancing entails transaction costs (Harvey et al., 2025), these costs should affect TDF

Thrift Savings Plan (TSP) for federal employees, TDF expense ratios are close to the weighted averages of their underlying fund fees, suggesting that it should be possible to provide TDFs without wrapper fees. Lastly, although integrated fund managers could use TDF wrapper fees to subsidize 401(k) recordkeeping services, offering TDFs are associated with only a modest offset in direct recordkeeping compensation. Taken together, these facts suggest that wrapper fees are unlikely to be driven primarily by the cost or value added of life-cycle rebalancing or recordkeeping.

Third, we show that flows into TDFs are significantly less sensitive to fees than flows into non-TDFs. This low fee sensitivity is consistent with the role of TDFs as default options and implies weaker incentives for TDF managers to compete on fees. In addition, there are two other potential sources of market imperfections that allow TDF managers to exploit default investors' low fee sensitivity. First, employer sponsors of 401(k) plans face agency frictions when choosing which TDF to offer as default option for their participants. Second, there are entry barriers in the TDF market, as many TDF managers provide 401(k) recordkeeping services and can influence employers' choice of default options.

Overall, these facts indicate that higher fees on TDFs likely reflect fund managers' ability to charge higher markups, suggesting that default options create market power and allow fund managers to price discriminate against fee-insensitive TDF investors.

In the second part of the paper, we develop and estimate an equilibrium model to quantify TDF market power and fund managers' incentive to price discriminate. In the model, fund managers sell mutual funds both directly in the broad mutual fund market and indirectly through their TDFs. Managers compete on fees in both markets to maximize total profits. We show that equilibrium wrapper fees depend on the difference in the (inverse) fee elasticities of TDF and non-TDF investors. Hence, fund managers charge higher wrapper fees to exploit fee-insensitive TDF investors.

To capture the sources of differences in fee sensitivities, we model four groups of agents, whose decisions jointly determine the demand for TDFs and non-TDFs. First, each employer sponsor of 401(k) plans chooses a TDF as the default option for their plan participants. Second, 401(k) participants choose whether to invest in the TDFs chosen by their employers. Third, among 401(k) participants who do not have access to TDFs or opt out of TDFs, we model their demand for non-TDF funds. Lastly, investors outside 401(k) plans allocate among non-TDFs.³

We estimate our structural model following the empirical IO literature, specifically [Berry \(1994\)](#) and [Berry et al. \(1995\)](#). Our estimates suggest large and significant differences in fee sensitivities between TDF and non-TDF investors. TDF investors, including employers, are largely insensitive to fees with elasticities less than 1. By contrast, non-TDF investors exhibit high fee-sensitivity with elasticities between 2 and 3. To address

returns and are typically not directly captured by expense ratios.

³TDF assets outside 401(k) plans are mostly held in IRA accounts, which are rolled over from 401(k) plans. So we assume the demand inside 401(k) plans captures these rolled-over assets.

potential endogeneity in fund fees, we implement a Hausman instrument, instrumenting fund fees with the average fee of funds in other investment categories offered by the same manager (Egan et al., 2021). We also estimate similar results using custodian and transfer agency fees as a cost-shifter instrument (Janssen and Thiel, 2025) and using product differentiation instruments (Gandhi and Houde, 2020).

Using these demand estimates, we invert fund managers' first-order conditions under Nash-Bertrand competition to recover markups. Consistent with the large difference in fee elasticities, we find that TDF markups are, on average, 20 bps higher than those of their underlying funds. In aggregate, TDF managers extract roughly \$2.8 billion profits in 2019.

Using model estimates, we evaluate three counterfactual policies designed to curb market power in the TDF market. The first policy prevents fund managers from price discriminating between TDF and non-TDF investors, imposing zero wrapper fees. In 2019, this policy reduces TDF fees by 13 bps and increases TDF investors' welfare by \$1.2 billion. In equilibrium, fund managers raise non-TDF fees to offset lost profit from TDFs. However, they only raise fees by 3 bps, thanks to higher elasticities and the larger share of non-TDF investors. As a result, fund managers' profits decrease by 1.3% while total investor surplus stays largely unchanged from the status quo.

The second policy imposes a 20 bps cap on TDF fees. The surplus of TDF investors increases by approximately \$1.9 billion, driven by an average decline of over 25 bps in TDF fees.⁴ The surplus of non-TDF investors remains nearly unchanged as the cap implicitly limits fund managers' ability to respond by raising fees on TDF underlying funds. Hence, these welfare gains come at a substantial cost to fund managers, whose profits fall by 5% relative to the status quo.

The last policy aligns employer and investor preferences by raising employers' fee sensitivity to that of non-TDF investors outside 401(k) plans. More elastic TDF demand pushes down TDF fees substantially by 27 bps, raising TDF investor welfare by \$1.8 billion. Fund managers' profits fall by over 3% relative to the status quo.

Our findings do not imply TDFs are undesirable investment products. To the contrary, previous evidence suggests that TDFs can substantially increase retirement saving by increasing equity exposure and reducing idiosyncratic risk. Mitchell and Utkus (2022) estimate that a low-cost TDF can increase an individual's retirement wealth by 50%. One caveat is that the TDF needs to be low-cost. Applying the same calculation, we find that a 60 bps wrapper fee (around the 90th percentile according to Table 1) would reduce the terminal wealth gain from 50% to 35%. In other words, while the investor would still be better off investing in the TDF, fund managers extract a sizable fraction of this gain through higher markups on TDFs.

Overall, our counterfactual policies address the following friction: TDFs have impor-

⁴This welfare change includes a modest welfare loss due to the exit of high-cost TDF managers, because we assume investors value the variety of TDFs

tant welfare implications for retirement savings but are sold by profit-maximizing fund managers to fee-insensitive investors. We show that regulating market power in the TDF market could yield substantial welfare gains for retirement investors.

Related Literature. This paper contributes to three strands of the literature.

First, it contributes to a growing literature on target-date funds (TDFs). One part of this literature studies how the adoption of TDFs affects 401(k) participants' investment behavior and retirement outcomes. TDF adoption increases equity exposure and reduces idiosyncratic risk (Mitchell and Utkus, 2022; Parker et al., 2025; Choukhmane and De Silva, 2026; Keim and Mitchell, 2017), as well as outperforming broker-recommended portfolios (Chalmers and Reuter, 2020). Another part of this literature examines TDFs' investment performance and cost structure. Several studies document systematic underperformance and attribute much of it to excessive fees (Sandhya, 2011; Brown and Davies, 2021; Massa et al., 2022), while Balduzzi and Reuter (2018) document heterogeneity in TDFs' ex-ante risk-taking, consistent with fund managers increasing risk exposure to attract flows following the 2006 PPA.⁵ This paper advances this literature by developing the first structural model of the TDF market, which allows us to quantify the extent of TDF managers' market power and to evaluate counterfactual policies designed to mitigate it.

Second, this paper contributes to the literature on default options in 401(k) plans.⁶ This literature shows that the adoption of defaults, combined with automatic enrollment, substantially increases participation and savings among workers (Madrian and Shea, 2001; Choi et al., 2004; Beshears et al., 2009), and that these effects are strongest upon adoption and at short horizons but attenuate over time (Beshears et al., 2021; Choukhmane, 2025; Choukhmane and Palmer, 2024; Choi et al., 2024). This paper complements this literature by focusing on the supply side of default provision. We show that fund managers exploit the role of TDFs as default options to sustain higher markups, and that constraining their market power can yield large welfare gains for 401(k) investors, in addition to the benefits from the adoption of TDFs as default investments.

Lastly, this paper contributes to a growing literature that applies empirical industrial organization methods to study retirement markets from a supply-side perspective. Motivated by empirical evidence on agency frictions in the design of 401(k) plans (Pool et al., 2016, 2025; Doellman and Sardarli, 2016; Badoer et al., 2020; Gropper, 2023), several recent papers develop structural models to quantify these frictions and assess the impact of policies aimed at reducing plan costs (Bhattacharya and Illanes, 2022; Loseto, 2023; Yang, 2023; Zhao, 2026). Related applications outside the U.S. include research on the

⁵There are also papers studying asset pricing implications of TDF rebalancing. Parker et al. (2023) show that contrarian flows of TDFs reduce stock returns and volatility. Fang and Goldstein (2025) show that rebalancing of TDFs increases stock-bond correlation.

⁶The introduction of behavioral nudges such as default options in 401(k) plans is motivated by a large body of evidence documenting suboptimal investment behavior among plan participants. See, for example, Benartzi (2001), Benartzi and Thaler (2001, 2007), Agnew et al. (2003), Poterba (2003), Choi et al. (2009), Tang et al. (2010), and Sialm et al. (2015).

Mexican pension system (Hastings et al., 2017) and the Chilean pension system (Illanes, 2017; Illanes and Padi, 2021; Boehm, 2024).

In addition, we are also related to studies of the industrial organization of mutual fund markets. An et al. (2023) analyze competition between index providers and ETFs. Roussanov et al. (2020) study how marketing expenses determine fund size. Brown et al. (2023) show how inertia drives ETF market power. Massa (2003) and Gavazza (2011) study the effects of mutual fund family product varieties and demand spillovers.

The remainder of the paper proceeds as follows. Section 2 describes the data sources. Section 3 presents motivating evidence on market power in the TDF market. Section 4 outlines the structural model. Section 5 presents the estimation results. Section 6 discusses the counterfactual policy analysis, and Section 7 concludes.

2 Target-Date Funds, Data, and Summary Statistics

2.1 Target-Date Funds

TDFs allocate across asset classes and rebalance from equities toward lower risk assets, such as bonds, as investors approach retirement. This process, known as the “glide path”, aligns portfolio risk with investors’ human capital and remaining investment horizon.

Rise of TDFs. Since the Pension Protection Act (PPA) of 2006, TDFs have become the dominant default investment options in employer-sponsored 401(k) plans. The PPA allowed TDFs to be designated as Qualified Default Investment Alternatives (QDIAs), enabling employers to automatically enroll participants into these funds while receiving fiduciary protection. Before the PPA, default investments were often limited to money market funds. Following the PPA in 2006, the demand for TDFs increased sharply. In 2024, total assets under management in TDFs reached \$4 trillion (Figure 1) and 87.2% of plans with a qualified default investment alternative (QDIA) choose a TDF as the default (PSCA, 2024).

Almost all TDF assets are held through retirement accounts, including defined contribution (DC) plans and individual retirement accounts (IRAs). TDF assets in IRAs mostly come from 401(k) rollovers. According to ICI (2026), over 95% of flows to IRAs come from rollovers in the past decades. Around 60% of TDF assets are held in target-date mutual funds, while the rest are in target-date collective investment trusts (CITs) as of 2019. The share of CITs has been rising, reaching around half in 2024. Because of limited regulatory disclosure requirements, we do not have detailed data on CITs. Hence, we focus our analysis on mutual fund TDFs.⁷

Fund-of-funds structure. TDFs are structured as funds-of-funds, with portfolios in-

⁷Unlike mutual funds, CITs are not registered with the Securities and Exchange Commission (SEC) and are exempt from the Investment Company Act of 1940. See Bollen et al. (2026); Tian and Shi (2024) for discussions on the rise of CITs

vested in a set of underlying funds offered by the same manager as the TDF itself.⁸ This structure is central to our analysis because it allows fund managers to sell the same underlying funds directly in the broad mutual fund market and indirectly through TDFs, creating scope for price discrimination between TDF and non-TDF investors. For CITs, that scope may be even greater because fund managers can set different fees on the same funds across 401(k) plans. The mechanism we study therefore extends to CITs.

Employers’ choice of TDFs. While participants in 401(k) plans invest in TDFs through their default options, their employers choose which TDF to offer with the help of financial service providers. These providers, known as *recordkeepers*, administer 401(k) plans, help design plan menus, and may also provide educational resources to participants. Many recordkeepers are also fund managers themselves (e.g., Vanguard and Fidelity).

Employers face fiduciary duties on behalf of participants when they select investment options in plan menus and choose recordkeepers. Even though the PPA in 2006 allows employers to use a TDF as the default option, the fiduciary standard still applies to employers’ choice of the specific TDF. However, in recent years, there have been a number of “excessive fee” lawsuits, in which participants accuse their employers of offering expensive funds, for example, a more expensive retail share class when the plan is eligible for a cheaper institutional share class of the same fund. Notably, *Hughes v. Northwestern* reached the U.S. Supreme Court in 2022. Despite their fiduciary duties, employers likely face various sources of agency costs (Pool et al., 2016, 2025; Doellman and Sardarli, 2016; Badoer et al., 2020; Gropper, 2023; Bhattacharya et al., 2024; Loseto, 2023; Yang, 2023) that could prevent them from sufficiently internalizing participants’ interests.

2.2 Data Sources

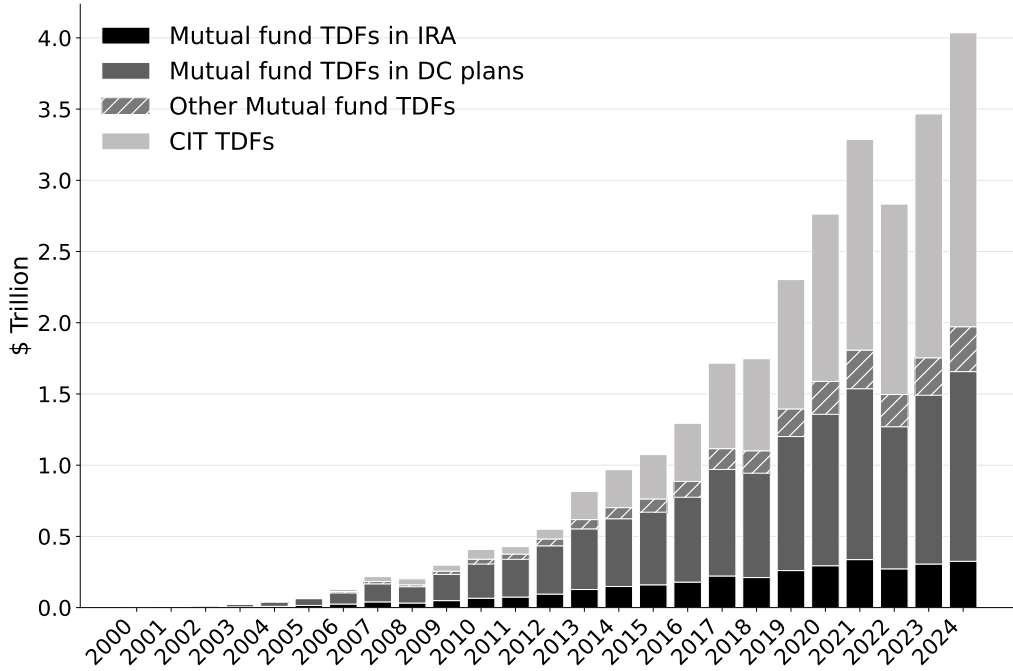
We combine 401(k) plan-level data from BrightScope Beacon with data on mutual funds and ETFs from CRSP and Morningstar Direct.

BrightScope Beacon provides detailed plan-level information for defined contribution plans, covering over 80% of plan assets. BrightScope collects the data from Form 5500, which employer sponsors file with the Department of Labor (DOL) every year. Our BrightScope sample is from 2009 to 2019. For each plan $\times$ year, BrightScope provides the menu of investment options and total amount allocated in each option across all participants. Specifically, the data allow us to observe whether employer sponsors offer any TDF to their participants, and which TDFs they offer. We merge our investment menu data from BrightScope with additional plan characteristics directly from Form 5500.

We obtain mutual fund (including ETF) assets under management, expense ratios, classifications, and fund managers from Morningstar Direct. We classify TDFs based on Morningstar *global category*. We use *fund ID* to identify share classes of the same fund,

⁸Except for a handful of small TDFs, nearly all underlying funds are proprietary funds managed by the TDF manager (Table 1).

Figure 1: TDF Asset Under Management



TDF AUM held through different investment vehicles. Source: Investment Company Institute (ICI).

and *strategy ID* to identify funds with different target retirement years of the same TDF series. Finally, we also obtain TDF holdings from CRSP.

When analyzing fees, we use mutual fund data from Morningstar and CRSP up to 2024. Because BrightScope data is up to 2019, we stop at 2019 for our structural estimation.

2.3 Default Option and TDF

Our data include an indicator for whether a plan offers a default option. However, we do not know which investment option is the default. Survey data suggest that the vast majority of 401(k) plans use a TDF as the default option. According to data from the PSCA 68th Annual 401(k) Survey, 87% of plans offer a TDF as the default option as of 2024 (PSCA, 2024). According to data from plans administered by Vanguard, 98% use a TDF as the default option (Vanguard, 2025). Therefore, we assume the largest TDF is the default option of the plan.

Due to data limitations, we do not separately analyze the effect of default options other than TDFs in this paper. Survey evidence discussed above suggests that non-TDF defaults are uncommon. Investors also appear to prefer TDFs as default options. For example, Choukhmane and De Silva (2026) show that 42% of investors opt out of a default money market fund, compared with only 5% who opt out of a default TDF. Therefore, we consider default options and TDFs as effectively equivalent in 401(k) plans for the

purposes of our analysis.

2.4 TDF Fees and Wrapper Fees

We define the TDF wrapper fee as the difference between a TDF’s expense ratio and that of a replicating portfolio based on its underlying funds.

We construct the replicating portfolio in three steps. First, for each TDF, we identify the prospectus-listed underlying funds using holdings data from CRSP. Second, we aggregate across share classes to obtain a fund-level measure of the expense ratios for both TDFs and underlying funds. Third, we replace each prospectus underlying fund with the cheapest fund offered by the same fund manager in the same Morningstar global category.

This approach captures two channels through which fund managers can increase TDF fees: directly, by charging a higher wrapper fee on top of the underlying funds, and indirectly, by selecting underlying funds with higher fees.⁹

We make two important remarks on our measure of wrapper fees. First, since 2017, several asset managers introduced “internal” underlying share classes with zero fees, without substantially reducing TDF expense ratios. These share classes are not available for direct purchase and can only be used as underlying investments in the asset manager’s proprietary funds. For example, in 2019, T. Rowe Price introduced a zero fee share class, Class Z, that is available only to funds managed by T. Rowe Price, including its TDFs. Because we want wrapper fees to measure the additional fees investors pay when they purchase the underlying funds through TDFs rather than directly in the mutual fund market, we exclude these internal share classes before aggregating to the fund level.

Second, our procedure for identifying accessible underlying funds considers only the cheapest funds offered by the *same* fund manager as the TDF itself. This restriction is relevant because we want wrapper fees to capture the extent to which fund managers price discriminate between TDF and non-TDF investors by selling the same set of funds.

Importantly, this restriction makes our measure of wrapper fees more conservative than alternative measures. For example, wrapper fees more than double when we relax the same-fund manager restriction and replace the prospectus underlying funds with the cheapest funds in the same category (Figure D.3). This is similar to the approach in [Brown and Davies \(2021\)](#), who replicate TDF underlying funds with a portfolio of low-cost Vanguard ETFs.

On the other hand, we can construct an even more conservative measure by conditioning on the same fund manager, category, and whether the fund is an index or active fund. We consider this alternative measure to be overly restrictive, because one way TDF managers could extract rents is to offer active funds for default investors even though cheaper

⁹To illustrate the indirect channel, we compare our baseline measure of wrapper fees with the wrapper fees directly constructed from the prospectus underlying funds in Figure B.2. Appendix B provides additional details on the wrapper fee construction.

Table 1: Summary Statistics

	Num Obs	Mean	St. Dev.	Pct10	Pct50	Pct90
<i>Employer Level</i>						
Offer TDF	36,555	0.84	0.37	0.00	1.00	1.00
Have Default Option	36,555	0.86	0.35	0.00	1.00	1.00
Have Auto-Enrollment	36,555	0.32	0.47	0.00	0.00	1.00
Allocation to TDF	34,036	0.41	0.26	0.08	0.38	0.80
Proprietary non-TDFs	24,043	0.27	0.28	0.00	0.16	0.68
Proprietary TDFs	21,304	0.38	0.48	0.00	0.00	1.00
<i>Fund Level Asset-weighted</i>						
<i>TDF</i>						
Expense Ratio (pp)	57	0.39	0.26	0.12	0.45	0.67
Wrapper Fee (pp)	57	0.22	0.20	0.08	0.10	0.59
Proprietary Underlying Funds	57	0.97	0.13	1.00	1.00	1.00
Number of Underlying Funds	57	12.29	7.36	4.21	17.52	19.49
Share in 401(k)	57	0.53	0.13	0.44	0.57	0.58
Manager is Recordkeeper	57	0.86	0.35	0.00	1.00	1.00
<i>Non-TDF</i>						
Expense Ratio (pp)	8,782	0.42	0.35	0.05	0.31	0.94
Share in 401(k)	8,782	0.07	0.11	0.00	0.02	0.25
Manager is Recordkeeper	8,782	0.45	0.50	0.00	0.00	1.00

Summary statistics as of 2019. The top panel shows statistics at the employer level, where we restrict to employers whose 401(k) plans have over 75% of assets invested in mutual funds. Allocation to TDF is only based on mutual fund TDFs. The bottom panel shows summary statistics at fund level. Statistics for TDFs are reported at the TDF series level. Expense ratios are winsorized at the 1% level.

index equivalents are available. Even with this restrictive measure, wrapper fees are only 5 bps lower, on average. This reduction is small compared to the typical difference between index and active fund fees. One reason is that many fund managers exclusively offer active funds. Figure D.4 shows the fraction of index funds among non-TDFs for the largest nine TDF managers as of 2019. Several fund managers, such as T. Rowe Price and Capital Group, offer almost no index funds at all.

Overall, our procedure yields a relatively conservative measure of the wrapper fee that captures the scope for price discrimination between TDF and non-TDF investors while ensuring that the underlying funds are accessible to non-TDF investors.

2.5 Summary Statistics

We report summary statistics as of 2019 in Table 1. The top panel is at the employer (plan) level, where we restrict to employers whose 401(k) plans have over 75% of assets invested in mutual funds.¹⁰ Over 80% of employers offer a TDF in their 401(k) plans as

¹⁰Our sample based on plans that largely invest in mutual funds cover 65% of plans and 82% of mutual fund TDF assets in the full BrightScope sample.

of 2019. 86% have default investment options and 32% have automatic enrollment. In 401(k) plans with mutual fund TDFs, participants allocate 40% of their assets to TDFs on average, consistent with TDFs being popular default investment options.

Employers work with financial service providers, known as recordkeepers, to design plan menus. Some of these recordkeepers are also fund managers. Among employers who work with such integrated recordkeepers, 27% of non-TDFs and 38% of TDFs are proprietary funds. The higher share of proprietary TDFs suggests a stronger incentive for fund managers to offer their TDFs as the default options in 401(k) plans.

In the bottom panel, we report fund-level summary statistics weighted by fund assets. Statistics for TDFs are reported at the TDF series level.¹¹ The average TDF expense ratio is 39 bps, while the average wrapper fee is 22 bps. Although some TDFs charge wrapper fees below 10 bps, TDFs at the 90th percentile charge around 60 bps above the fees of their underlying funds.

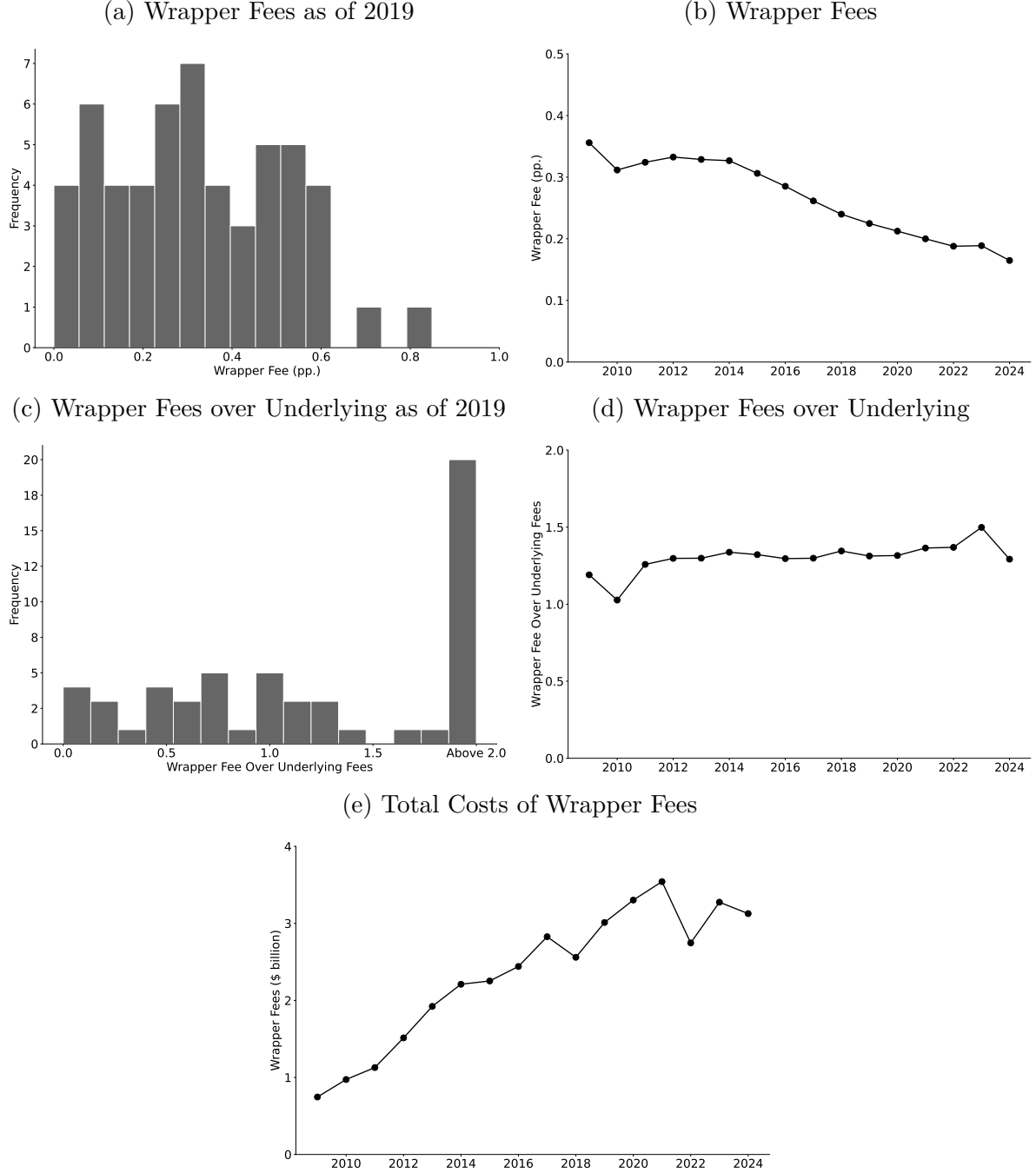
With the exception of a handful of small TDFs, nearly all underlying funds of a TDF are proprietary funds of the TDF manager itself. Most TDFs hold between 5 and 20 underlying funds.

Non-TDFs include both funds we classify as underlying funds of TDFs and other non-underlying funds. The average non-TDF expense ratio is 42 bps. Comparing TDFs vs non-TDFs, 53% of TDF assets come from 401(k) plans, whereas only 7% of non-TDF assets come from 401(k) plans. According to [ICI \(2025\)](#), in 2019, 67% of mutual fund TDF assets come from DC plans with the rest mostly from IRAs. Our 401(k) share is lower due to sample coverage of BrightScope data. Lastly, TDF managers are also much more likely to provide recordkeeping services.

3 Motivating Evidence

We present three sets of facts that highlight the potential market power of TDF managers. First, we show that TDFs charge substantial wrapper fees relative to the fees of their underlying funds. Second, we present evidence indicating that the additional costs of managing TDFs or the value added from TDF rebalancing are unlikely to explain wrapper fees. Third, we show that TDF flows are less fee-sensitive than non-TDF flows. Taken together, these findings suggest that TDF managers exploit fee-insensitive default investors by charging higher fees on TDFs than on the underlying funds.

Figure 2: TDF Wrapper Fees and Underlying Fees



Panel (a) plots the distribution of wrapper fees as of 2019. Wrapper fees are constructed based on the baseline definition described in Section 2.4. Panel (b) plots asset-weighted average TDF wrapper fee over time. Panel (c) plots the distribution of the ratio of wrapper fees over underlying fees as of 2019. Panel (d) plots the asset-weighted average ratio of wrapper fees over underlying fees. Panel (e) plots total costs of wrapper fees for TDF investors.

3.1 TDF Wrapper Fees

We begin by showing that TDFs charge higher fees than portfolios replicating their underlying holdings. Exploiting the fund-of-funds structure, we compare TDF expense ratios with the asset-weighted expense ratios of their underlying funds and compute TDF wrapper fees using the procedure described in Section 2.4.

Figure 2a plots the distribution of wrapper fees across TDFs as of 2019. Similar to the summary statistics in Table 1, TDFs charge substantial wrapper fees above the fees of their underlying funds. The dispersion of wrapper fees is also large. While some TDFs charge almost zero wrapper fees, others charge over 60 bps more than the fees of the underlying funds. Figure 2c plots the distribution of wrapper fees as a fraction of underlying fund fees. Many TDFs charge wrapper fees that are over twice as large as those of the underlying funds.

Figure 2b shows that wrapper fees averaged around 30 bps in 2010 and declined to roughly 20 bps over the sample period. This pattern is consistent with the broader secular decline in mutual fund fees. Appendix Figure D.2 shows that both total TDF fees and underlying fund fees declined over the same period. However, Figure 2d indicates that wrapper fees remain above 100% of underlying fund fees, suggesting that fund managers continue to earn substantially higher fees from TDFs relative to their underlying funds. With the growth of TDF assets, these wrapper fees have translated into sizable costs for investors, exceeding \$3 billion annually over the past decade (Figure 2e).¹²

Figure 2 is based on our baseline definition of wrapper fees, where we identify a TDF’s underlying funds as the cheapest accessible funds within the same fund manager $\times$ Morningstar global category cohort as the corresponding prospectus underlying fund. Figure D.3 shows that wrapper fees are roughly 70% higher when we identify underlying fund without conditioning on the same manager. In 2019, this alternative wrapper fee implies over \$5 billion in costs to TDF investors.

In Figure 3, we compare wrapper fees and underlying fund fees for several large TDFs. Figure 3a corresponds to the T. Rowe Price Retirement 2030 Fund. It charges total fees ranging from about 60 to 80 bps over our sample period. Such high fees reflect both substantial wrapper fees often above 30 bps and also relatively high fees of active underlying funds.¹³

Figure 3b and Figure 3c correspond to two Fidelity Freedom 2040 funds, the active TDF, Fidelity Freedom 2040 Fund, and the index counterpart, Fidelity Freedom Index 2040 Fund. Since we replicate these TDFs using the cheapest Fidelity funds among

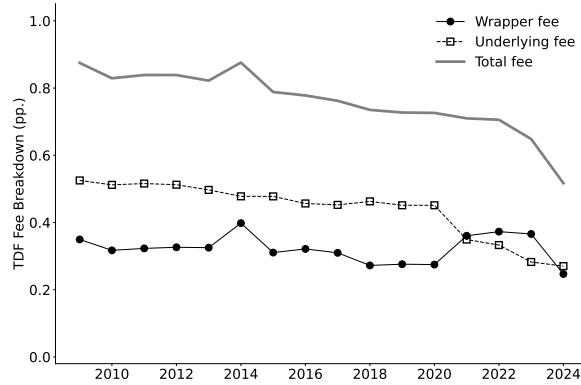
¹¹A TDF series is a set of TDFs, offered by the same fund manager, with different target retirement years. For example, Fidelity offers the Fidelity Freedom TDF series, which contains several TDFs: the Fidelity Freedom 2020 for individuals who are expected to retire around 2020, the Fidelity Freedom 2030 for those expected to retire around 2030, etc.

¹²These revenues are based on mutual fund TDFs only, excluding TDFs structured as CITs.

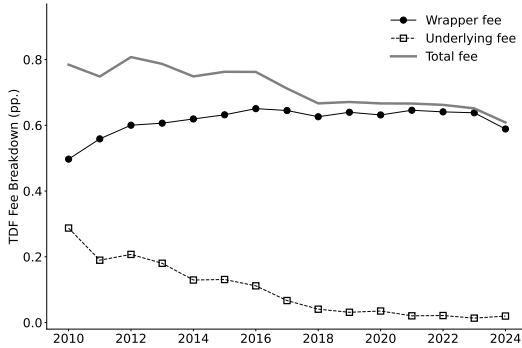
¹³Figure D.4 shows that T. Rowe Price has very few index funds among its non-target-date funds. Hence, the underlying funds are mostly active funds with relatively high fees.

Figure 3: Wrapper Fee Examples

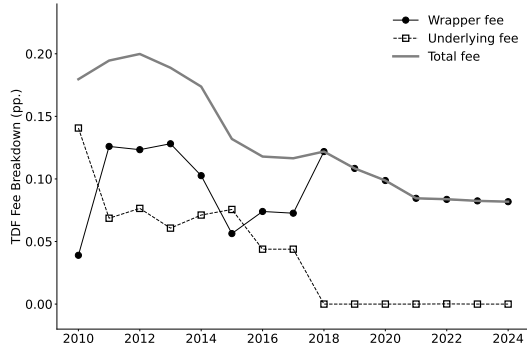
(a) T. Rowe Target Retirement 2030



(b) Fidelity Freedom 2040



(c) Fidelity Freedom Index 2040



Panel (a) plots TDF fee, underlying fee, and wrapper fee breakdown for T. Rowe Price Retirement 2030 Fund. Panel (b) plots TDF fee, underlying fee, and wrapper fee for the Fidelity Freedom 2040 fund. Panel (c) plots TDF fee, underlying fee, and wrapper fee for the Fidelity Freedom Index 2040 fund. Figure D.5 shows more examples.

their categories, the underlying funds are similar whenever index funds are available. Underlying fund fees decline sharply during the sample period toward nearly zero from 2018 onward. Despite the decline in underlying fees, wrapper fees on these two TDFs actually increase over time, from 50 to over 60 bps for the active TDF and from 5 to 10 bps for the index TDF. While the underlying funds face intense competition in the index fund market, it appears that wrapper fees can insulate Fidelity TDFs from this competitive pressure.

Overall, these patterns suggest that TDF managers are able to maintain higher fees on TDFs and that reductions in underlying fund fees are not fully passed through to TDF investors.

3.2 Cost and Value Added Explanations of Wrapper Fees

A natural interpretation of wrapper fees is that they compensate fund managers for the incremental costs or value added associated with TDFs, including the life-cycle rebalancing and also 401(k) plan servicing. Although these costs and quality are not directly observ-

able, several pieces of evidence suggest that they are unlikely to explain the magnitude of observed wrapper fees. We provide a summary and leave the details to Appendix C.

First, the main incremental portfolio management service provided by a TDF is the life-cycle rebalancing. This service is operationally simple and highly scalable for professional fund managers. To benchmark the cost of such rule-based portfolio management, we compare TDF wrapper fees with the expense ratios of index funds, which also implement rebalancing strategies. Index funds typically rebalance over more underlying securities than TDFs, yet their expense ratios are substantially lower than TDF wrapper fees, according to Figure C.1. In 2019, the asset-weighted expense ratio of index funds is 7 bps, compared with an average TDF wrapper fee of 22 bps. This comparison suggests that direct management costs are unlikely to explain wrapper fees.

Although life-cycle rebalancing may be valuable to investors, the scope for persistent rents from providing this service should be limited. Large asset managers can implement glide paths efficiently, so in a competitive TDF market any value created by life-cycle rebalancing should be competed away. The comparison with index funds is again instructive: index funds also add value by pooling and rebalancing broad portfolios of securities, yet they charge relatively low fees.

Second, TDF managers may create value by optimizing rebalancing in ways that reduce transaction costs (Harvey et al., 2025). For example, Vanguard (2024) estimates transaction cost savings of roughly 10 bps. These transaction costs, however, should affect fund performance rather than expense ratios and are therefore conceptually distinct from the wrapper fee.¹⁴ Moreover, other large TDF managers presumably also optimize rebalancing, so any rents associated with this value added should again dissipate in a competitive market.

Third, the Thrift Savings Plan (TSP) for federal employees provides a useful benchmark for the cost or value added of offering TDFs at scale. TSP investors can choose among a small set of index funds and TDFs that invest in these funds. According to Table C.1, TSP TDF expense ratios are very close to the weighted average of the expense ratios of their underlying funds, implying little or no wrapper fee. This suggests that it is feasible to operate TDFs without charging a substantial additional layer of fees.

Finally, wrapper fees may also indirectly compensate fund managers for providing 401(k) recordkeeping services. Integrated fund managers earn fees both through fund expense ratios and through direct recordkeeping compensation. However, 401(k) plans with proprietary TDFs have only modestly lower direct recordkeeping fees around 2 to 3 bps based on estimates reported in Table C.2. The implied recordkeeping fee offset is small relative to the size of wrapper fees. Recordkeeping costs therefore explain at most a limited portion of observed wrapper fees.

¹⁴The prospectus of the Vanguard Target Retirement Fund states: “The Fund pays transaction costs, such as purchase fees, when it buys and sells securities... These costs, which are not reflected in annual fund operating expenses or in the example, affect the Fund’s performance.”

Table 2: Flow-Fee Sensitivity

	Fund Flow (%)	
Δ Expense Ratio (%)	-0.723*** (0.032)	-0.666*** (0.032)
Δ Expense Ratio $\times$ TDF (%)	0.570** (0.226)	0.481** (0.224)
N	108,059	107,927
R2	0.14	0.21
Global Category $\times$ Index $\times$ Year Fixed Effects		X

Fund-flow sensitivity to fees. Estimates are based on Equation (1). We winsorize flow and percentage fee changes at the 5% level. For non-TDFs, we focus on equity, fixed income, and allocation funds, which are comparable to underlying funds held by TDFs. Our sample ranges from 2010 to 2019.

Taken together, these facts suggest that wrapper fees are unlikely to be driven by either the incremental costs or the incremental quality of life-cycle rebalancing and recordkeeping. The persistence of substantial wrapper fees instead suggests that TDF managers are able to charge higher markups on TDFs than on the underlying funds, consistent with lower fee sensitivity among default investors.

3.3 Fee Elasticity

If wrapper fees do not reflect the costs of managing life-cycle rebalancing, they are likely to capture differences in demand elasticities between TDF and non-TDF investors. To examine whether TDF investors are less sensitive to fees than investors in comparable non-TDFs, we estimate fee elasticities by regressing fund flows on percentage changes in expense ratios using the following specification:

$$Flow_{it} = \eta \times \% \Delta fee_{it} + \eta^{tdf} \times \% \Delta fee_{it} \times 1\{TDF_i\} + X_{it}\beta + \gamma_{it} + \epsilon_{it}. \quad (1)$$

We compute fund flows as the percentage change in total net assets (TNA) net of returns, $Flow_{it} = TNA_{it}/TNA_{it-1} - Ret_{it}$. Here, η captures the elasticity of flows with respect to fees, and η^{tdf} captures the difference in elasticities between TDFs and non-TDFs. X_{it} includes controls for fund age and lagged returns. To capture potentially non-linear relationships between returns and flows, we control for third-order polynomials of fund returns over the past year, separately for TDFs and non-TDFs. γ_{it} are asset category $\times$ index $\times$ year fixed effects that control for fund heterogeneity.

Table 2 shows that non-TDF funds have an estimated flow elasticity of about -0.7. The coefficient η^{tdf} is positive and statistically significant, implying that the elasticity of TDFs is much lower. This gap in fee sensitivity is consistent with the role of TDFs as default options. Investors who are automatically enrolled into TDFs, or who leave rebalancing to the predetermined glide path, are less likely to actively compare fees across alternatives.

By contrast, investors who actively choose non-TDFs have greater fee sensitivities.

From the perspective of fund managers, this difference in elasticities creates scope for price discrimination, where they charge higher markups on TDFs to exploit the lower fee sensitivities of default investors.

3.4 Other Frictions in TDF Markets

In addition to the low fee sensitivity of default TDF investors, at least two other frictions in the TDF market allow fund managers to price discriminate. First, employer sponsors of 401(k) plans may not fully offset the low fee sensitivity of default investors. When selecting the TDF as the default option, employers face a range of agency frictions, including search costs, switching costs, and misaligned incentives (Bhattacharya et al., 2024; Loseto, 2023; Yang, 2023). As a result, they often permit recordkeepers to include expensive funds in the plan menu (Pool et al., 2016, 2025; Doellman and Sardarli, 2016). The impact of these frictions is also reflected in the recent wave of litigation over high fee investment options in 401(k) plans.

Second, the TDF market has entry barriers. Most large TDF managers also provide 401(k) recordkeeping services, through which they can influence employer sponsors' choice of default options. Without control over the distribution channel, it can be hard to generate sufficient take-up among default investors. Appendix Figure D.6 shows that, despite the wave of entry around the Pension Protection Act of 2006, the numbers of TDF managers and TDF strategies have declined since 2018. For example, Goldman Sachs liquidated its TDFs as of 2022.

Relatedly, Table 1 shows that when employers work with recordkeepers who are also fund managers, 27% of non-TDFs and 38% of TDFs offered on plan menus are proprietary funds of the recordkeepers. Such high percentages highlight both agency frictions, where employers potentially allow fund managers to take advantage of their recordkeeping role to market their own funds, and also barriers to entry for fund managers who do not provide recordkeeping services.

Next, we introduce a structural model to capture differences in fee-sensitivity between TDF investors (including both employers and 401(k) participants) and non-TDF investors, in a market of imperfect competition among fund managers.

4 Model

We present a model of fee competition between differentiated fund managers who sell their products in two markets: directly in the mutual fund market and indirectly by packaging them into TDFs.¹⁵ Such market segmentation allows fund managers to price

¹⁵We assume all underlying funds of TDFs come from the same fund managers. In our data, around 95% of TDF assets belong to the same fund managers, which is similar to statistics reported in Brown and Davies (2021). Hence, we consider our assumption to be relatively innocuous.

discriminate, charging TDF wrapper fees on top of the fees of underlying funds.

To capture the drivers of different fee sensitivities, we separately model employer sponsors of 401(k) plans, 401(k) investors who choose default options and those who actively invest, and lastly investors outside 401(k) plans. The decisions of these groups of agents jointly determine the demand for TDFs and non-TDFs. Let $\mathcal{J}$ and $\mathcal{D}$ collect the set of non-TDFs and TDFs. Let $\mathcal{J}_f$ and $\mathcal{D}_f$ denote the set of non-TDFs and TDFs offered by fund manager f . The model captures the following sequence of choices:

1. Each fund manager f sets the fees of their non-TDFs $j \in \mathcal{J}_f$ and fees of their TDFs $k \in \mathcal{D}_f$, which together determine the wrapper fees.
2. Each employer sponsor m chooses whether to offer a TDF $k \in \mathcal{D}$ in their 401(k) plan, or not to offer one.
3. Each investor i in plan m makes an allocation decision in two steps:
 - 3a. The investor decides whether to invest in the TDF k offered in plan m .
 - 3b. The investor who opts out of the TDF (or if none is offered by the employer) allocates among non-TDFs in $\mathcal{J}$ that are offered among 401(k) menus.
4. An investor i outside 401(k) plans allocates among non-TDFs in $\mathcal{J}$.

In step 2, we consider the TDF as the “default option”. In practice, this default can take several forms: it may serve as the default for automatically enrolled participants, as the default for participants who voluntarily enroll, or as an “informal default” where the TDF is recommended to participants. For tractability, we abstract from modeling different default features and instead represent them through the employer’s decision to offer a TDF.¹⁶

Step 3a captures how 401(k) participants choose whether to remain in the default option or to opt out and make their own investment decisions. We do not model a joint portfolio allocation problem over both TDFs and other funds on the plan menu. Although investors in plans with more attractive non-TDF options may be more likely to opt out of the default, most default investors are unlikely in practice to be fully aware of the non-TDF options available on the menu when making that decision. In addition, recent evidence shows that most TDF investors do not hold other funds (Mitchell and Utkus, 2022; Vanguard, 2025).¹⁷ We therefore treat Steps 3a and 3b as separate decisions. We discuss robustness to this separation assumption in our estimation in Section 5.2.

¹⁶We consider TDFs at the *strategy* level. We consider TDFs with different retirement years of the same strategy as one fund. Employers rarely offer more than one TDF strategy. In rare cases with multiple TDF strategies, we designate the TDF strategy with larger total AUM as the chosen TDF.

¹⁷According to Mitchell and Utkus (2022), 64% of investors who use TDFs invest exclusively in TDFs. Among TDF investors in plans administered by Vanguard, 71% held only a single TDF in 2024 (Vanguard, 2025).

Step 3b captures both how the menu of non-default mutual funds is determined and how investors allocate across non-TDFs available on their plan menus. Because our focus is the TDF market, we abstract from modeling the full design of 401(k) plan menus for tractability.

Step 4 assumes that investors outside 401(k) plans choose only among non-TDFs. We do not model demand for TDFs outside 401(k) plans because most TDF assets held outside 401(k) plans are in IRAs, which are largely funded through 401(k) rollovers. For the median traditional IRA account, 401(k) rollovers account for 95% of flows and 80% of assets (ICI, 2024), suggesting that demand for TDFs within 401(k) plans provides a reasonable approximation to TDF demand outside 401(k) plans.

4.1 Employer TDF Choice (Step 2)

Employer m chooses a TDF $k \in \mathcal{D}$ or no TDF. We assume employer m derives the following utility from offering TDF k :

$$u_{kt}^{emp} + \epsilon_{mkt} = -\alpha^{emp} p_k + X'_{kt} \beta^{emp} + \epsilon_{mkt},$$

where p_k is the expense ratio of TDF k . α^{emp} captures employers' fee sensitivities, which can account for their agency frictions. X_{kt} includes fund-level controls for fund age, index funds, an indicator for whether the manager of k is also a recordkeeper, and year fixed effects. ϵ_{mkt} captures idiosyncratic preferences that follow the Type I extreme value distribution.

We normalize the utility of the outside option of no TDF to zero. The outside option includes both the case where the employer offers no default and the case where the employer offers other types of default such as money market funds. As discussed earlier, we focus on TDFs and do not distinguish among alternative default options.

The probability that employer m selects TDF k is the following.

$$q_{kt}^{emp} = \frac{\exp(u_{kt}^{emp})}{1 + \sum_{k' \in \mathcal{D}} \exp(u_{k't}^{emp})}. \quad (2)$$

4.2 Investors in 401(k) Plans (Step 3)

401(k) investor i first makes a binary decision of whether to invest in the TDF k chosen by her employer m (Step 3a). We assume at this step, investor i is unaware of the rest of the plan menu, capturing the role of TDFs as default options.¹⁸ The utility from investing

¹⁸Effectively, this is a discrete choice decision model with imperfect consideration. Demand models where consumers do not consider all options available in the market have a long-standing tradition in the empirical industrial organization literature. See, for example, Goeree (2008), Ho, Hogan and Scott Morton (2017), and Abaluck and Adams-Prassl (2021).

in TDF k is

$$u_{mkt}^{tdf} + \epsilon_{imkt} = -\alpha^{tdf} p_k + X'_{mkt} \beta^{tdf} + \epsilon_{imkt},$$

where X_{mkt} includes fund-level controls, plan-level characteristics, such as whether the employer offers a default option or automatic enrollment, and year fixed effects.

Across investors in employer m 's plan, the fraction of assets in plan m allocated to TDF k is

$$q_{mkt}^{tdf} = \frac{\exp(u_{mkt}^{tdf})}{1 + \exp(u_{mkt}^{tdf})}. \quad (3)$$

Next (Step 3b), 401(k) investors who opt out of TDFs or whose employers do not offer TDFs choose among the remaining funds $\mathcal{J}$. We do not model the specific menu design of each plan and instead allow investors to choose among all funds offered in 401(k) plans. For allocations among non-TDFs, the utility of fund j is

$$u_{jt}^{in} + \epsilon_{ijt} = -\alpha^{in} p_j + X'_{jt} \beta^{in} + \xi_{jt}^{in} + \epsilon_{ijt}. \quad (4)$$

Since investors at the stage is more likely to be informed about fund characteristics, we allow investors to have different fee sensitivities when they allocate among non-TDFs (α^{in}) than when they decide whether to opt out of default TDFs (α^{tdf}). Because we do not model employers' menu choice, α^{in} captures the fee sensitivities of both employers and 401(k) participants. X_{jt} includes a rich set of controls with fixed effects at the fund investment category $\times$ index $\times$ year level.

Aggregating across 401(k) investors, the fraction of assets allocated to fund $j \in \mathcal{J}$ is the following. We consider allocation to money market funds as the outside option and normalize its utility to zero.

$$q_{jt}^{in} = \frac{\exp(u_{jt}^{in})}{1 + \sum_{l \in \mathcal{J}} \exp(u_{lt}^{in})}. \quad (5)$$

4.3 Investors Outside 401(k) Plans (Step 4)

Investors outside 401(k) allocate among non-TDFs in $\mathcal{J}$. Outside investors can choose among all funds in $\mathcal{J}$, including mutual funds and investment vehicles not commonly available in 401(k) plans, such as ETFs. The share of outside 401(k) assets allocated to fund j is denoted by q_{jt}^{out} and is analogous to Equation (5).

While the allocation problem is similar for the two groups of non-TDF investors, we estimate their preferences separately, where outside investors have fee coefficient α^{out} . Identification relies on different sources of variation in observed fund market shares: variation within 401(k) plans identifies the preferences of 401(k) investors, while variation in the broader mutual fund market identifies the preferences of outside investors.

4.4 Fund Manager Fee-Setting Problem (Step 1)

Fund managers compete in a standard Nash-Bertrand fee-setting game. Again, $\mathcal{J}_f \subseteq \mathcal{J}$ and $\mathcal{D}_f \subseteq \mathcal{D}$ denote, respectively, the sets of non-TDFs and TDFs offered by manager f . Manager f chooses fees to maximize total profit:

$$\max_{(p_j, p_k)_{j \in \mathcal{J}_f, k \in \mathcal{D}_f}} \sum_{j \in \mathcal{J}_f} (p_j - c_j) D_j^a(p) + \sum_{k \in \mathcal{D}_f} (p_k - c_k) D_k^d(p),$$

where $D_j^a(p)$ is the demand for non-TDF j and $D_k^d(p)$ is the demand for TDF k . Since investors inside 401(k) plans can allocate across TDFs and non-TDFs, their demand functions depend on the entire vector of fees in $\mathcal{J} \cup \mathcal{D}$. For simplicity, we drop the time subscript.

Because we abstract from modeling TDF demand outside 401(k), the demand for TDF $k \in \mathcal{D}$ is

$$D_k^d = \underbrace{\sum_m M_m q_{km}^{emp} q_{km}^{tdf}}_{401(k) \text{ demand}}$$

where M_m denotes assets in plan m (including assets rolled over to IRAs), and M_{out} denotes total assets outside 401(k) plans.

The expression of the demand for non-TDF $j \in \mathcal{J}$ instead includes both demand from 401(k) and outside 401(k) investors:

$$D_j^a = \underbrace{\sum_m M_m \left(1 - \sum_{k \in \mathcal{D}} q_{km}^{emp} q_{km}^{tdf} \right) q_j^{in}}_{401(k) \text{ demand}} + \underbrace{M_{out} q_j^{out}}_{\text{outside 401(k) demand}}$$

In a Nash-Bertrand equilibrium, all fund providers set fees simultaneously to maximize profits taking competitors' fees as given. The resulting equilibrium markups are characterized by the following system of $|\mathcal{J}| + |\mathcal{D}|$ equations:

$$\begin{bmatrix} p^a - c^a \\ p^d - c^d \end{bmatrix} = - \begin{bmatrix} \Delta_{aa} & O \\ \Delta_{da} & \Delta_{dd} \end{bmatrix}^{-1} \begin{bmatrix} D^a \\ D^d \end{bmatrix} \quad (6)$$

where p^a is the $|\mathcal{J}|$ -vector of non-TDF fees, p^d is the $|\mathcal{D}|$ -vector of TDF fees, and Δ is a $|\mathcal{J}| + |\mathcal{D}|$ by $|\mathcal{J}| + |\mathcal{D}|$ matrix defined as

$$\Delta = \begin{bmatrix} \Delta_{aa} & O \\ \Delta_{da} & \Delta_{dd} \end{bmatrix} \equiv \nabla D \odot \Omega = \begin{bmatrix} \nabla_a D^a \odot \Omega^a & O \\ \nabla_d D^a \odot \Omega^a & \nabla_d D^d \odot \Omega^d \end{bmatrix}$$

where ∇D is the demand Jacobian and Ω is the product ownership matrix. Note that since 401(k) investors choose among non-TDFs after opting out of default TDFs, fees of

non-TDFs do not affect demand for TDFs ($\Delta_{ad} = O$).

We can decompose TDF fees into wrapper fees plus underlying fees as the following, where W is a $|\mathcal{D}| \times |\mathcal{J}|$ matrix collecting the holding shares of each TDF

$$p^d = \underbrace{\tilde{p}}_{\text{wrapper fee}} + \underbrace{Wp^a}_{\text{underlying fee}}$$

Assuming $c^d = Wc^a$,¹⁹ the equilibrium $|\mathcal{D}|$ -vector of wrapper fees is given by

$$\tilde{p} = -\Delta_{dd}^{-1}D^d - \left(-W\Delta_{aa}^{-1} + \Delta_{dd}^{-1}\Delta_{da}\Delta_{aa}^{-1}\right)D^a \quad (7)$$

We provide detailed derivations in Appendix A.

We can further simplify the expression under the special case where TDF and non-TDF markets are fully segmented, so TDF fees also do not affect the demand for non-TDFs ($\Delta_{da} = O$). In addition, suppose each fund manager holds one TDF and one underlying fund with $W = I$. Then, Equation (7) simplifies to:

$$\tilde{p} = (-\Delta_{dd}^{-1}D^d) - (-\Delta_{aa}^{-1}D^a) = \underbrace{\left(-\frac{D^d}{\partial D^d / \partial p^d}\right)}_{\text{inverse semi-elasticities of TDF}} - \underbrace{\left(-\frac{D^a}{\partial D^a / \partial p^a}\right)}_{\text{inverse semi-elasticities of non-TDF}} \quad (8)$$

where the optimal wrapper fee depends on the difference of (inverse) semi-elasticities between TDF and non-TDF markets. Equation (8) captures the economics of managers' fee-setting behavior. The magnitude of wrapper fees increases in the difference in fee sensitivities between default TDF investors and those who invest in non-TDFs.

5 Estimation Results

We estimate four demand systems: (i) employers' demand for TDFs, (ii) 401(k) investors' demand for TDFs chosen by their employers, (iii) demand for non-TDFs among 401(k) investors, and (iv) demand for non-TDFs among investors outside 401(k) plans.

To account for heterogeneity across employers, we estimate the model separately for four groups of employers based on their number of participants (<200, 200-500, 500-1000, >1000). Larger employers are potentially more sophisticated and face a greater threat of lawsuit. We also allow 401(k) investors to be heterogeneous across these four groups of employers. These investors may also have different preferences or sophistication. Outside 401(k) plans, we estimate only one demand system, since we do not have information to further segment investors outside 401(k).

To address the endogeneity of fees, we follow Egan, MacKay and Yang (2021) and construct Hausman instruments, defined as the average fees of funds managed by the

¹⁹This assumption is made only for expositional purposes and is not applied in estimation and counterfactuals.

Table 3: Employer Demand Estimates

	Employer Default Choice							
	<200		200-499		500-999		>1000	
	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV
Expense Ratio (pp)	-1.19*** (0.02)	-1.93*** (0.02)	-1.38*** (0.02)	-1.96*** (0.02)	-1.64*** (0.02)	-2.12*** (0.03)	-2.19*** (0.02)	-2.52*** (0.02)
Recordkeepers	1.73*** (0.01)	1.33*** (0.01)	1.61*** (0.01)	1.16*** (0.01)	1.52*** (0.02)	0.93*** (0.02)	1.55*** (0.02)	0.81*** (0.02)
log(Fund Age)	1.70*** (0.01)	0.98*** (0.02)	1.93*** (0.01)	1.22*** (0.02)	2.13*** (0.02)	1.60*** (0.02)	2.60*** (0.02)	2.10*** (0.02)
Past Returns (pp)	0.13*** (0.00)	0.12*** (0.00)	0.13*** (0.00)	0.12*** (0.00)	0.15*** (0.00)	0.13*** (0.00)	0.12*** (0.00)	0.12*** (0.00)
N	2,071,364	2,071,364	1,951,474	1,951,474	769,609	769,609	972,400	972,400
Elasticity	-0.42	-0.68	-0.50	-0.71	-0.59	-0.77	-0.74	-0.86
Year FE	X	X	X	X	X	X	X	X

MLE estimates of Equation (2). We restrict to plans with over 75% of assets invested in mutual funds, and drop plans whose TDF is not a mutual fund.

same fund manager in other Morningstar global categories. To avoid capturing joint fee-setting decisions, we exclude TDFs and manager $\times$ Morningstar category cohorts where any funds are held by TDFs when computing average fees. The exclusion restriction requires the average fees of other funds from the same manager to be uncorrelated with a fund's unobserved demand. This restriction is plausible if the average fees of other funds primarily proxy for the manager's operating costs. When part of these costs is common across funds from the same manager, for example because of scale or scope economies, the Hausman instrument also satisfies the relevance condition.

In Appendix D, we repeat part of our analysis using two different types of instruments: cost-shifter instruments based on fund-level custodian and transfer agency fees, and product differentiation instruments. Custodian and transfer agency fees capture operational costs in safekeeping the fund assets, maintaining shareholder records, and handling client communications. The exclusion restriction requires such costs to be uncorrelated with investors' unobserved demand. This assumption is plausible if, after controlling for observed fund characteristics such as fund age and asset class, variation in custodian and transfer agency fees primarily reflects these operational costs. The relevance condition holds whenever some of these costs are passed through to investors via fund fees. Overall, the results are qualitatively similar to the baseline results with the Hausman instrument.

The product differentiation instruments are constructed following [Gandhi and Houde \(2020\)](#). These instruments are constructed from each product's position in the characteristics space relative to rival products using sums of pairwise squared differences in characteristics, capturing how isolated or crowded a product is relative to its substitutes. The exclusion restriction is that, conditional on own-product characteristics, rival characteristics are orthogonal to the product's unobserved demand. As for the cost-shifter instruments, we obtain similar results when we use these differentiation instruments.

5.1 Employer TDF Choice

We estimate employers’ TDF choice at the plan level using maximum likelihood (MLE). To address fee endogeneity, we introduce a control function with the Hausman instrument. We proceed in two steps. First, we regress TDF expense ratios on the instruments and controls. Second, we include a polynomial of the first-stage residuals in the MLE estimation.

Table 3 reports the estimates. We estimate a more negative fee coefficient α^{emp} when using control functions with our Hausman instrument. The implied elasticities to TDF fees are slightly below one, suggesting that employers are relatively insensitive to fees. Employers’ low fee sensitivities could reflect low financial sophistication, agency frictions, or low valuation for plan menu quality (Bhattacharya et al., 2024; Loseto, 2023; Yang, 2023). Employers also prefer TDFs with longer track records, higher past returns, and TDFs whose managers are also recordkeepers.

On a relative basis, large employers appear to be more sensitive to fees, consistent with a higher level of sophistication or a greater threat of lawsuits. They also place lower value on integrated TDF managers that also serve as recordkeepers, suggesting that they are less likely to allow their recordkeepers to designate their own TDFs as default options without scrutiny. Table D.2 reports estimates based on cost-shifter and product differentiation instruments. We estimate slightly larger but qualitatively similar elasticities.

5.2 401(k) Investor TDF Choice

We estimate 401(k) investors’ decision to invest in the employer-selected TDF using Equation (3) following Berry (1994). Table 4 reports our estimates at the plan-year level. We estimate small α^{tdf} across all four employer size groups, which is not surprising since investors may choose the default options without knowing their expense ratios. Even with instruments, fee coefficients are only slightly negative and our estimates imply elasticities close to zero. Table D.3 reports similar estimates using cost-shifter or product differentiation instruments.

We also find that investors allocate more to TDFs in plans with formal default options or automatic enrollment. Investors are more likely to invest in TDFs with longer track records and managed by integrated managers. Investors do not appear to value past returns, which is also consistent with choosing the default options without learning about their past performance.

At this stage, we assume that investors are unaware of non-TDF options. If, instead, investors choose TDFs after comparing them with non-TDFs, ignoring non-TDFs could introduce omitted-variable bias and potentially attenuate our fee coefficient estimates. As a robustness check, we control for the inclusive value of non-TDFs on their plan menus, which effectively allows us to model how investors choose among both TDF and non-TDF

Table 4: Demand Estimates for 401(k) TDF Investors

	401(k) Investor Allocation to Default							
	OLS	<200 Hausman IV	OLS	200-499 Hausman IV	OLS	500-999 Hausman IV	OLS	>1000 Hausman IV
Expense Ratio (pp)	0.03 (0.03)	-0.18*** (0.03)	-0.03 (0.02)	-0.27*** (0.03)	-0.18*** (0.03)	-0.44*** (0.04)	-0.04 (0.03)	-0.18*** (0.03)
log(Fund Age)	0.62*** (0.03)	0.64*** (0.03)	0.72*** (0.03)	0.75*** (0.03)	0.63*** (0.04)	0.69*** (0.04)	0.27*** (0.04)	0.32*** (0.04)
Default Option	0.43*** (0.02)	0.43*** (0.02)	0.49*** (0.02)	0.50*** (0.02)	0.39*** (0.03)	0.40*** (0.03)	0.29*** (0.02)	0.29*** (0.02)
Auto-Enroll	0.34*** (0.02)	0.34*** (0.02)	0.39*** (0.01)	0.39*** (0.01)	0.43*** (0.02)	0.42*** (0.02)	0.37*** (0.02)	0.37*** (0.02)
Recordkeepers	0.25*** (0.02)	0.21*** (0.02)	0.14*** (0.02)	0.08*** (0.02)	0.10*** (0.03)	0.03 (0.03)	0.08*** (0.03)	0.03 (0.03)
Past Returns (pp)	-0.03*** (0.00)	-0.02*** (0.00)	-0.01* (0.00)	-0.00 (0.00)	0.01 (0.01)	0.01** (0.01)	0.01** (0.00)	0.02*** (0.01)
log(Plan Partcp)	0.44*** (0.03)	0.44*** (0.03)	0.40*** (0.02)	0.40*** (0.02)	0.44*** (0.04)	0.45*** (0.04)	0.31*** (0.01)	0.31*** (0.01)
log(Plan Assets)	-0.40*** (0.01)	-0.40*** (0.01)	-0.40*** (0.01)	-0.40*** (0.01)	-0.39*** (0.01)	-0.40*** (0.01)	-0.37*** (0.01)	-0.38*** (0.01)
N	70,403	70,403	72,803	72,803	32,864	32,864	42,758	42,758
Elasticity	0.01	-0.05	-0.01	-0.07	-0.05	-0.12	-0.01	-0.05
Year FE	X	X	X	X	X	X	X	X

Estimates of Equation (3) at the plan-year level. We restrict the sample to plans that offer mutual fund TDFs.

options. Table D.5 shows that we estimate similar fee elasticities after controlling for the inclusive value. In addition, while we expect the coefficient in front of the inclusive value to be negative if investors choose from both TDF and non-TDFs, our estimates are close to zero. Therefore, it appears that investors do not consider non-TDFs when they choose whether to allocate to default TDFs.

5.3 401(k) Investors' Allocation Among Non-TDFs

Next, we estimate the demand among investors who allocate among non-TDFs in 401(k) plans. We aggregate allocation across 401(k) plans, capturing the joint decisions of how employers determine non-TDF options on 401(k) menus and how 401(k) investors allocate among those options. Again, we estimate Equation (5) following Berry (1994).

We report the estimates in Table 5 under “Inside 401(k)”. When allocating assets among non-TDFs, 401(k) investors are estimated to be much more sensitive to fees than investors who default into TDFs, with elasticities between 2 and 3. Investors appear to value funds with longer track records, higher past returns, and whose manager is also a recordkeeper.

5.4 Allocation of Investors Outside 401(k) Plans

Lastly, we estimate demand for investors outside 401(k) plans. We assume investors outside 401(k) choose only among non-TDFs. The vast majority of TDF assets are either held in 401(k) plans or rolled over to IRAs from 401(k) plans. So our TDF demand within

Table 5: Demand Estimates for Non-TDF Investors

	Inside 401(k)										Outside 401(k)	
	<200		200-499		500-999		>1000		Outside		Outside	
	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV
Expense Ratio (pp)	-1.58*** (0.05)	-4.30*** (0.16)	-1.72*** (0.05)	-3.96*** (0.17)	-1.90*** (0.06)	-3.47*** (0.18)	-2.65*** (0.06)	-4.76*** (0.21)	-1.48*** (0.02)	-4.20*** (0.09)		
log(Fund Age)	1.52*** (0.02)	1.56*** (0.02)	1.59*** (0.02)	1.63*** (0.03)	1.52*** (0.03)	1.56*** (0.03)	1.58*** (0.03)	1.61*** (0.03)	1.31*** (0.01)	1.32*** (0.01)		
Recordkeepers	1.30*** (0.04)	0.62*** (0.05)	1.41*** (0.04)	0.86*** (0.05)	1.41*** (0.04)	1.04*** (0.06)	1.88*** (0.04)	1.39*** (0.06)	0.95*** (0.02)	0.19*** (0.04)		
Past Returns (pp)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.03*** (0.00)	0.03*** (0.00)	0.03*** (0.00)	0.02*** (0.00)		
N	29,893	29,893	29,563	29,563	24,817	24,817	27,512	27,512	56,349	56,349		
Elasticity	-1.01	-2.76	-1.08	-2.48	-1.14	-2.09	-1.39	-2.50	-0.85	-2.43		
Category $\times$ Year $\times$ Index FE	X	X	X	X	X	X	X	X	X	X		

Estimates of Equation (4) at the fund-year level with year $\times$ Morningstar category $\times$ index fixed effects. In columns (1) to (8) under “Inside 401(k)”, we estimate demand across funds in 401(k) plans. In the last two columns under “Outside 401(k)”, we estimate demand across funds outside 401(k) plans.

401(k) should already capture TDF demand in IRAs.

According to estimates in Table 5 under “Outside 401(k)”, investors are again much more fee-sensitive than TDF investors, with an elasticity of 2.4. Compared to non-TDF investors inside 401(k) plans, outside investors place lower value on recordkeepers, consistent with integrated managers having a weaker influence outside 401(k). The coefficient is still positive, plausibly because recordkeepers tend to be well-known fund managers and carry certain brand value.

Table D.4 reports the demand estimates for non-TDF investors when we use cost-shifter and product differentiation instruments. Despite noisier estimates, the results are broadly aligned with our baseline estimates using the Hausman instrument. Elasticities for non-TDF investors are higher than elasticities of both employers and TDF investors in 401(k) plans.

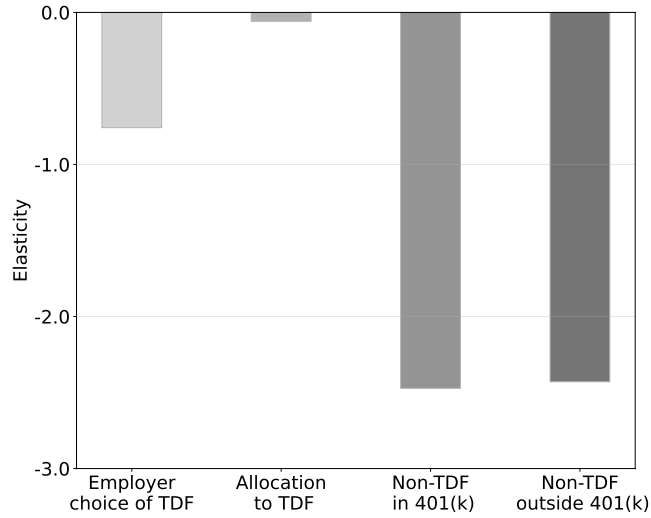
5.5 Comparison of Elasticities

As a summary, Figure 4 reports the estimated demand elasticities for the four demand systems. The demand for TDFs, both employers’ choice of TDFs as the default options and 401(k) investors’ decision to invest in default options, are inelastic to fees. The magnitude of both elasticities is on average below one and nearly zero for workers. The low elasticity of plan participants is consistent with default options, while the low elasticity of employers could reflect low financial sophistication or agency frictions.

Conversely, non-TDF investors are much more elastic to fees with elasticities between 2 and 3 (in absolute terms) for both investors inside and outside 401(k). Through the lens of our model, this large gap in elasticities gives fund managers incentives to price discriminate between TDF and non-TDF investors.

Our elasticities are comparable to those found in similar settings. Brown et al. (2023) estimate elasticities between 1.3 and 2.8 in demand for index funds. Baker et al. (2022) estimate similar elasticities between 1.5 and 3.4 in demand for ESG funds. In pension

Figure 4: Estimated Demand Elasticities



Estimated elasticities to fees, based on estimates using Hausman instrument. Estimates are reported both for TDF investors, separately for employer and investors, and non-TDF investors within and outside 401(k). We report weighted average across employer size groups.

fund choices in Mexico’s privatized social security system, [Hastings et al. \(2017\)](#) estimate elasticities ranging between 0.3 and 1. Counterfactual elasticities without sales force range between 1.25 and 3. The effect of sales force is similar to the difference in elasticities between TDF and non-TDF investors in our setting.

In demand for other financial products, researchers have estimated low elasticities of 0.2-0.75 in bank deposit markets ([Dick, 2008](#); [Egan et al., 2017](#); [Xiao, 2020](#)) and 0.47 in equity brokerage markets ([Di Maggio et al., 2022](#)). On the other hand, others have estimated relatively higher elasticities of 2.18 in life insurance ([Koijen and Yogo, 2016](#)) and 2-6 in mortgage markets ([Buchak et al., 2024](#); [Robles-Garcia, 2019](#); [Benetton, 2021](#)). Our estimates of elasticities of TDF and non-TDF investors broadly fall within this range.

5.6 Supply Side: Markups

We use our demand estimates to recover fund managers’ markups by inverting the Nash–Bertrand equilibrium conditions in Equation (6). For simplicity, we assume that each fund manager offers up to three composite funds: a TDF, an underlying fund held by the TDF, and a non-underlying fund. Underlying funds are based on cheapest funds within each manager $\times$ Morningstar global category cohort based on our wrapper fee procedure in Section 2.4. We also include funds whose fees are within 10% of the lowest fees in their cohorts. These funds are likely to be close substitutes to the cheapest funds in their cohorts, and their fees may have similar responses in our counterfactuals.

The first panel of Table 6 reports markups based on our estimates using the Hausman instrument for the year 2019. Our estimates suggest that TDF markups are roughly 20

bps higher than markups on their underlying funds. As assets in TDF mutual funds (excluding TDFs structured as CITs) reached \$1.4 trillion in 2019, our estimates imply that TDF managers generate about \$2.8 billion in profits from wrapper fees.²⁰

Our analysis focuses on potential price discrimination between TDFs and their underlying funds, while taking the fee dispersion among non-target-date mutual funds as given. Such dispersion may reflect investors' search costs (Hortaçsu and Syverson, 2004) or inertia (Brown et al., 2023), which can generate fee differences between underlying and non-underlying non-TDFs.

6 Counterfactuals

We use our model to evaluate three counterfactual policies aimed at limiting the market power of TDFs. The first policy requires fund managers to set identical fees for their TDFs and underlying funds. This rule eliminates TDF wrapper fees, effectively preventing price discrimination between TDF and non-TDF investors. The second policy imposes a cap on TDF fees to protect investors against market power. The third policy targets employers' agency costs, by aligning employers' fee sensitivities to those of investors outside 401(k) plans.

For each policy, we report counterfactual market shares, fees, and markups in Table 6. We also compute changes in investor surplus, TDF fees and fund manager profits relative to the status quo in Table 7. We separately report surplus for three groups of investors: TDF investors, 401(k) non-TDF investors, and investors outside 401(k). For TDF investors, we use the logit inclusive value from the employer's TDF choice problem in Equation (2), assuming employers make decisions representing 401(k) investors' interests. Because employers potentially have lower fee sensitivities due to agency frictions, we use the fee sensitivities of investors outside 401(k) when computing this inclusive value.²¹ For the other two groups of non-TDF investors, we use the inclusive value from their respective decisions. For each group, we compute their surplus as their respective inclusive value multiplied by their total assets.

6.1 No Wrapper Fee

First, we prohibit TDF managers from charging wrapper fees, effectively banning price discrimination between TDF and non-TDF investors. Restrictions on potentially unfair

²⁰The asset-weighted average marginal costs (differences between fees and markups) of TDFs and their underlying funds are slightly negative, mostly driven by low-cost index funds with large assets under management. These marginal costs may be negative because fund managers receive benefits beyond the stated expense ratios, including cross-selling opportunities for higher-fee products and revenues from securities lending.

²¹Using non-TDF 401(k) investors' fee sensitivity delivers similar results. We prefer to use the fee sensitivity of investors outside 401(k) because our estimated α^{in} could still reflect agency costs as it combines both employers' menu choice and participants' allocation decisions.

Table 6: Counterfactual Equilibrium Outcomes

	TDF	Underlying	Non-underlying	Total
<i>Status quo</i>				
Market share	4.59	20.16	58.80	83.54
Fee	0.41	0.18	0.52	0.43
Fee (unweighted)	0.51	0.27	0.60	0.51
Markups	0.48	0.28	0.27	0.29
<i>No wrapper fee</i>				
Market share	4.88	18.51	59.83	83.22
Fee	0.27	0.21	0.52	0.43
Fee (unweighted)	0.35	0.35	0.60	0.53
Markups	0.33	0.30	0.27	0.28
<i>TDF fee cap</i>				
Market share	5.07	20.02	58.49	83.58
Fee	0.16	0.18	0.52	0.41
Fee (unweighted)	0.17	0.27	0.60	0.49
Markups	0.21	0.28	0.27	0.27
<i>Align preferences</i>				
Market share	4.84	20.05	58.67	83.56
Fee	0.14	0.18	0.52	0.41
Fee (unweighted)	0.31	0.27	0.60	0.50
Markups	0.30	0.28	0.27	0.28

Market shares, average fee and average markup in status quo and counterfactuals. Outcomes are based on Hausman IV estimates for the year 2019. Marginal costs are calibrated after inverting price-setting equilibrium conditions in Equation (6). Market shares, fees, and markups are measured in percentage points. Column “total” reports total share and weighted average results across the three fund categories.

price-discrimination across consumers are common in many settings. For example, the Consumer Financial Protection Bureau (CFPB) prohibits mortgage loan originator compensation based on loan terms to prevent steering consumers toward expensive loans.²²

The second panel of Table 6 presents equilibrium outcomes under the no-wrapper-fee policy. Relative to the status quo, eliminating wrapper fees reduces TDF fees while increasing fees on the underlying funds. TDF fees fall by 15 and 13 bps respectively on an equal-weighted and asset-weighted basis. Thanks to lower fees, TDF market share increases slightly from 4.5% to 4.9%. The modest increase is consistent with TDF investors' low fee sensitivity.

By contrast, fees on underlying funds increase by 7 bps on an equal-weighted basis and by 3 bps on an asset-weighted basis. Because underlying funds represent a larger segment of the market and face more fee-sensitive investors, their fee increase is smaller than the corresponding fee reduction for TDFs. Nonetheless, this modest increase leads to a sizable decline of nearly 2% in the market share of underlying funds, as fee-sensitive non-TDF investors substitute toward non-underlying funds. In the aggregate, however, overall mutual fund fees and markups remain approximately unchanged relative to the status quo.

The first column of Table 7 reports the implied changes in investor surplus, and fund manager profits. Preventing fund managers from price discriminating substantially increases surplus for TDF investors by approximately \$1.2 billion. In response, managers recoup some of these losses by raising fees on underlying funds, which hurts non-TDF investors' welfare. However, their ability to do so is limited by the high fee sensitivity of non-TDF investors. As a result, fund manager profits decline by approximately 1.3% and total investor surplus stays roughly unchanged from the status quo.

Overall, this policy increases the surplus of TDF investors at the expense of non-TDF investors, while fund manager profits reduce slightly. While we do not model other supply-side responses, such as product repositions or exits, we consider such responses to be unlikely given the modest reduction in fund managers' profits.

6.2 Fee Cap

Second, we impose a cap on TDF fees. Fee caps are common price regulations to curb market power.²³ For example, the default options of certain occupational pension schemes are capped at 75 bps in the UK.²⁴ However, this cap is likely not binding and much higher than fees of portfolio replicating underlying funds.

In our counterfactual, we cap TDF fees at 20 bps, which is around the average fees of underlying funds under the status quo. We implement the fee cap in three steps. First,

²²<https://www.govinfo.gov/app/details/FR-2010-09-24/2010-22161>

²³Cuesta and Sepúlveda (2021) study the effect of interest rate cap on the consumer credit market in Chile.

²⁴<https://www.gov.uk/government/publications/the-charge-cap-guidance-for-trustees-and-managers-of-occupational-schemes>

we let TDFs with marginal costs exceeding the cap exit the market. Second, we solve for the equilibrium among the remaining TDFs. Finally, we cap optimal fees and re-solve for the equilibrium among managers for whom the cap does not bind.

The third panel of Table 6 reports the resulting equilibrium outcomes. Average asset-weighted TDF fees fall sharply under this policy by 25 bps relative to the status quo, a reduction of 60% from the status quo. Thanks to reduced TDF fees, TDF market share increases from 4.6% to 5.1%. Outcomes for non-TDFs remain mostly unchanged as the fee cap limits fund managers’ ability to increase fees on underlying funds in response.

The second column of Table 7 shows that the fee cap has larger effects on aggregate market outcomes. The surplus for TDF investors increases by \$1.9 billion.²⁵ However, this gain comes at the expense of fund managers whose profits reduce by over 5%.

6.3 Align Preferences

Third, we align employers’ preferences with those of outside 401(k) investors. This counterfactual essentially aims at mitigating employer agency costs, and one can think of employers in this counterfactual to be highly sophisticated who sufficiently internalize the welfare of their participants.

To implement this counterfactual, we resolve the equilibrium fees after replacing employers’ fee sensitivities α^{emp} with outside investors’ fee sensitivities α^{out} . Table 6 shows that TDF fees fall substantially by 27 bps weighted by assets and by 20 bps on an equal-weighted basis. The larger asset-weighted fee reduction indicates that employers with higher fee sensitivities substitute more toward low-cost TDFs. Outcomes of non-TDFs remain roughly the same as the status quo.

As higher fee sensitivities of employers push down TDF fees, TDF investors’ welfare increases by \$1.8 billion. The welfare of other investors remains roughly the same as the status quo, while fund managers’ profits fall by over 3%. This counterfactual suggests that if employer sponsors are sufficiently fee sensitive when they designate default options, TDF managers would not be able to extract more rent from default investors. In practice, changing employers’ preference can be difficult. The pressure of recent “excessive fee” lawsuits can push some employers to be more fee sensitive, but such effects may not extend to small or medium-sized employers with lower threat of lawsuits.

6.4 Discussion

Overall, our counterfactual policies increase the welfare of TDF investors, at the expense of non-TDF investors under the no-wrapper-fee policy and of fund managers under the fee cap and aligned preferences. Because these changes do not constitute a Pareto improvement, whether these outcomes are desirable depends on the weights a social planner

²⁵Note that this increase in welfare per bps of TDF fee reduction is lower than the no-wrapper-fee counterfactual after accounting for the loss of variety from TDF exits.

Table 7: Counterfactual Welfare Changes

	No wrapper fee	TDF fee cap	Align preferences
<i>Δ Consumer surplus</i>			
TDF	1.16	1.86	1.77
Non-TDF in 401(k)	-0.26	0.01	-0.00
Non-TDF outside 401(k)	-1.00	0.05	-0.03
Total	-0.11	1.92	1.74
<i>TDF Fees</i>			
Δ Fees	-0.13	-0.25	-0.27
% Δ Fees	-33.28	-60.69	-66.26
<i>Profits</i>			
Δ Profits	-0.76	-3.10	-1.99
% Δ Profits	-1.29	-5.23	-3.36

Counterfactual changes in consumer surplus, TDF fees, and profits relative to the status quo. Changes in consumer surplus and profits are in billion dollars. Changes in TDF fees are in percentage points. Outcomes are based on Hausman IV estimates for the year 2019.

assigns to the surplus of TDF investors, non-TDF investors, and fund managers. We argue that the planner may place greater weight on the welfare of TDF investors, because they may be less financially sophisticated and inadequate retirement savings can generate broader societal costs. From this perspective, the “friction” we address is that a highly welfare-relevant investment decision, the choice of TDF, is made by relatively fee-insensitive investors. Our counterfactuals mitigate this friction either by leveraging the decisions of fee-sensitive investors through a ban on price discrimination, directly regulating TDF fees, or reducing the extent of agency frictions in plan design.

7 Conclusion

This paper examines how default investment options generate market power in the U.S. retirement market. Using a structural model of the TDF market, we show that the interaction between investors’ low fee sensitivities and fund managers’ pricing incentives can sustain large markups on default investment products.

Empirically, we find that TDF investors are substantially less fee-sensitive than other mutual fund investors, and that TDF markups exceed those of comparable funds by about 20 bps. The resulting aggregate profit from wrapper fees amounts to roughly \$2.8 billion as of 2019.

Our counterfactual analysis shows that constraining fund managers’ ability to price discriminate between TDF and non-TDF investors could improve TDF investor welfare. Imposing no wrapper fees raises TDF investors’ surplus by nearly \$1.2 billion, with negligible effects on fund managers’ profits but a welfare loss of similar magnitude among non-TDF investors. On the other hand, a cap on TDF fees delivers even larger welfare gains of \$1.9 billion for TDF investors but entails a sharper reduction in managers’ prof-

its. Transferring surplus from non-TDF investors or fund managers to TDF investors may be desirable if the social planner places greater weight on the welfare of retirement savers who can be less financially sophisticated.

While TDFs can substantially increase retirement wealth, our results show that part of these gains may be captured by TDF managers who exploit fee-insensitive default investors. A simple calculation based on [Mitchell and Utkus \(2022\)](#) suggests that a 60 bps wrapper fee can reduce the retirement-wealth gain of TDF from 50% to 35%.

These findings highlight an important policy tension: while default options such as TDFs have been highly effective in improving retirement savings, investors' inertia and inattention can give rise to market power that erodes some of these gains. Designing regulation that preserves the behavioral benefits of defaults while limiting fund managers' ability to extract rents is therefore essential for improving the efficiency of the defined-contribution retirement system.

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Online Appendix

A Derivations

Equilibrium wrapper fee. We start from the expression for the equilibrium price-cost margins,

$$\begin{bmatrix} p^a - c^a \\ p^d - c^d \end{bmatrix} = - \begin{bmatrix} \Delta_{aa} & O \\ \Delta_{da} & \Delta_{dd} \end{bmatrix}^{-1} \begin{bmatrix} D^a \\ D^d \end{bmatrix}$$

and rewrite it as

$$\begin{bmatrix} p^a - c^a \\ p^d - c^d \end{bmatrix} = - \begin{bmatrix} \Delta_{aa}^{-1} & O \\ -\Delta_{dd}^{-1} \Delta_{da} \Delta_{aa}^{-1} & \Delta_{dd}^{-1} \end{bmatrix} \begin{bmatrix} D^a \\ D^d \end{bmatrix},$$

Letting $p^d = \tilde{p} + W(p^a - c^a)$, then the wrapper fee equals

$$\tilde{p} = -\Delta_{dd}^{-1} D^d + \Delta_{dd}^{-1} \Delta_{da} \Delta_{aa}^{-1} D^a - W(p^a - c^a).$$

Substituting for $(p^a - c^a)$ we obtain the expression in the main text.

Derivation of demand Jacobian. We derive the expression for the derivatives of the demand function for a default and non-default fund respectively.

Recall the demand for TDF $k \in \mathcal{D}$:

$$D_k^d = \sum_m M_m q_{km}^{emp} q_{km}^{tdf}$$

The derivative of D_k^d with respect to fees of TDF $l \in \mathcal{D}$ is given by

$$\frac{\partial D_k^d}{\partial p_l} = \sum_m M_m \left(\frac{\partial q_{km}^{emp}}{\partial p_l} q_{km}^{tdf} + q_{km}^{emp} \frac{\partial q_{km}^{tdf}}{\partial p_l} 1\{l = k\} \right)$$

The demand for non-TDF fund $j \in \mathcal{J}$ is

$$D_j^a = \underbrace{\sum_m M_m \left(1 - \sum_{k \in \mathcal{D}} q_{km}^{emp} q_{km}^{tdf} \right) q_j^{in}}_{401(k) \text{ demand}} + \underbrace{M_{out} q_j^{out}}_{\text{outside 401(k) demand}}$$

The derivative of non-TDF fund j with respect to fees of TDF $l \in \mathcal{D}$ reads

$$\frac{\partial D_j^a}{\partial p_l} = \sum_m M_m \left[\left(1 - \sum_k q_{km}^{emp} q_{km}^{tdf} \right) \frac{\partial q_j^{in}}{\partial p_l} - \left(\sum_k \frac{\partial q_{km}^{emp}}{\partial p_l} q_k^{tdf} + q_{km}^{emp} \frac{\partial q_k^{tdf}}{\partial p_l} 1\{k=l\} \right) q_j^{in} \right]$$

Fees of non-TDF funds will not affect demand for TDF funds. As such the derivative of D_k^d with respect to non-TDF $l \in \mathcal{J}$ is zero. The derivative of D_j^a demand with respect to non-TDF l is instead given by:

$$\frac{\partial D_j^a}{\partial p_l} = \sum_m M_m \left(1 - \sum_{k \in \mathcal{D}} q_{km}^{emp} q_{km}^{tdf} \right) \frac{\partial q_j^{in}}{\partial p_l} + M_{out} \frac{\partial q_j^{out}}{\partial p_l}.$$

B Data Construction

B.1 Data sources and TDF sample construction

We combine four data sources: Morningstar Direct, CRSP mutual fund summary data, CRSP mutual fund holdings, and BrightScope 401(k) plan filings. Morningstar Direct is our main source for expense ratios, fee components, fund net assets, fund categories, index-fund status, management company names, and target-date strategy identifiers. CRSP provides TDF portfolio identifiers and TDF portfolio holdings. BrightScope provides plan menus, balances, and plan characteristics.

We begin with the Morningstar annual share-class panel and define the target-date fund (TDF) universe as funds whose Morningstar category contains “Target-Date”. We keep observations from 2000 to 2024 with a non-missing CUSIP. We use CUSIP to merge Morningstar fund characteristics into CRSP summary and holdings data. From the CRSP summary data, we retain observations whose calendar date lies between the share class’s first offer date and end date, we collapse the quarterly summary data to one observation per CUSIP-year using the last available quarter of the year, and we filter for TDFs using the Morningstar TDF indicator. The resulting dataset contains CUSIP $\times$ year observations for TDFs and includes fund tickers, several fund characteristics from Morningstar and, importantly, the TDF portfolio identifier *crsp_portno* needed for the CRSP holdings data.

B.2 CRSP holdings and identification of underlying funds

For each TDF portfolio identified in the previous step, we retrieve all CRSP holdings with strictly positive portfolio weights and keep holdings reports through 2024. In the wrapper-fee construction, we use the last available holdings report in each calendar year so that the portfolio weights reflect the most recent annual composition of the TDF.

A small number of TDF holdings (less than 2.5% of observations) are reported without an identifier because the position corresponds to a master fund or another security reported under a nonstandard name.²⁶ For these observations, we use a hand-collected crosswalk based on the reported security name and then recover the missing underlying-fund CUSIP from Morningstar ticker–CUSIP matches in the corresponding year. For example, BlackRock Diversified Master Fund does not have an identifier. We match it with BlackRock Diversified Equity Fund with ticker BDVEX.

After linking holdings to Morningstar, we compute the dollar amount invested in each underlying fund as the product of the holding weight and TDF net assets. Aggregating these positions yields the assets held by TDFs in each underlying fund.

²⁶A master fund is the common underlying portfolio in a master fund structure, with one or more feeder funds investing all or nearly all of their assets in it.

B.3 Measuring TDF total fees

Morningstar Direct provides two expense-ratio variables. The first is a time-invariant snapshot variable, the *prospectus net expense ratio*, which records the fee reported in the prospectus at the time we extract the Morningstar data. The second is a time-varying variable, the *net expense ratio*, which is the series that is consistent with CRSP. For the vast majority of funds these two variables coincide at a given date. For TDFs, however, they can differ because TDFs are fund-of-funds.

After cross-checking annual reports and fund prospectuses from fund websites, we confirm that the Morningstar prospectus net expense ratio is the total fee paid by TDF investors. For most TDFs, the sum of the time-varying Morningstar net expense ratio and the portfolio-weighted expense ratio of the underlying funds equals the prospectus net expense ratio observed at the same date. We therefore use the time-varying Morningstar net expense ratio as the wrapper layer and add the underlying-fund layer to construct total TDF fees.

For a small set of fund series, Morningstar reports prospectus expense ratios that already coincide with the time-varying net expense ratio. Based on manual checks of prospectuses and fee histories, we treat these cases as reporting formats in which the underlying-fund layer is already embedded in the reported TDF net expense ratio. For those series, we therefore use the reported TDF net expense ratio itself as the total fee.

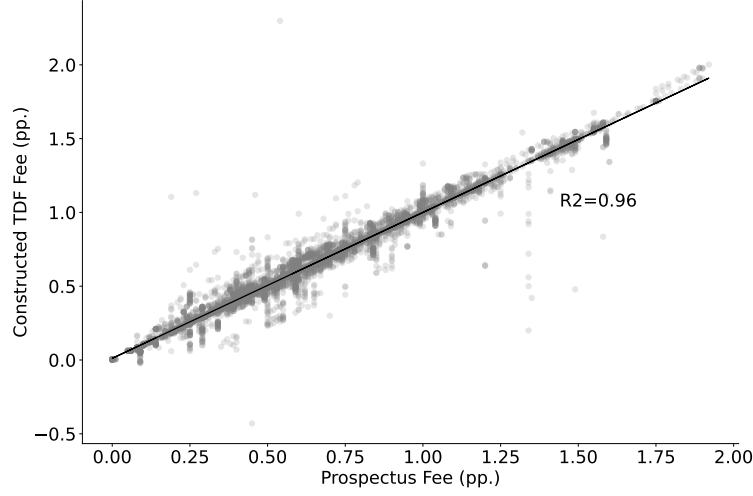
For completeness, Figure B.1 presents a scatter plot of the TDF total fee constructed as the sum of the TDF net expense ratio and the weighted average expense ratio of the TDF's underlying funds against the prospectus expense ratio. The R^2 of the fitted line is 96%, indicating that our construction closely tracks the total fee reported in the prospectus.

B.4 Measuring TDF underlying fees

Our goal is to construct a measure of wrapper fees that robustly captures the market power of TDF managers. The fund-of-funds structure of TDFs allows managers to make TDFs more expensive in two distinct ways. First, they can raise TDF fees *directly* by adding an explicit fee on top of the fees charged by the underlying funds (i.e., the wrapper fee). Second, they can raise TDF fees *indirectly* by selecting relatively expensive underlying funds.

To capture both margins, we replace the expense ratio of each underlying fund with the fee of the lowest-cost *accessible* comparable fund. A key issue is that some large fund families introduced zero- or near-zero-fee internal share classes that are used inside proprietary TDFs but are not meaningful outside options for investors purchasing funds directly. To ensure that the underlying-fee benchmark reflects fees available outside the TDF, we exclude these internal share classes when constructing the comparison set of accessible underlying funds.

Figure B.1: TDF total fees vs. TDF prospectus expense ratio



Scatter plot of TDF total fees vs. TDF prospectus expense ratio as of 2022. TDF total fees are constructed as the sum of the Morningstar net expense ratio and the average expense ratio of the TDF underlying funds.

Our baseline comparison set is formed within fund manager $\times$ Morningstar Global Category. In other words, for each underlying holding we search for the cheapest accessible comparable fund offered by the same manager and with the same broad investment style. Although TDFs typically hold institutional share classes of the underlying funds, we implement the comparison at the fund level rather than at the share class level. This allows us to account for the fact that smaller TDF investors may not be able to directly access the cheapest share classes of the underlying funds, thereby reducing the risk of overstating TDF wrapper fees.

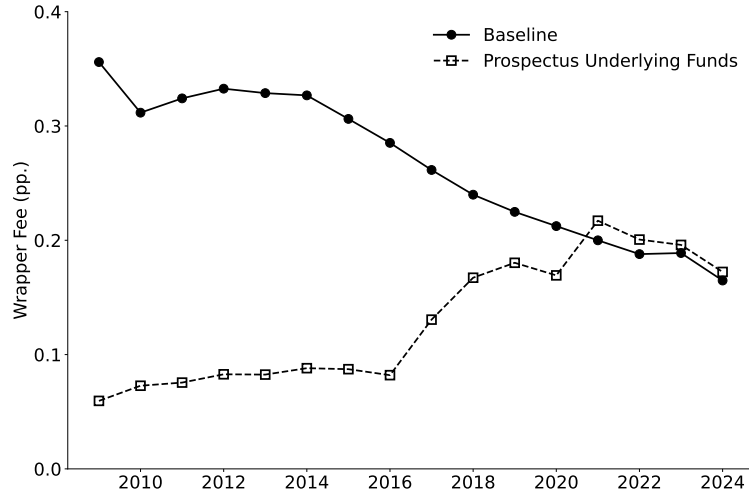
Overall, this procedure is consistent with our interpretation of wrapper fees. First, it captures the possibility that fund managers do not select the lowest-cost option among otherwise similar funds. Second, because TDFs invest in proprietary funds of the TDF manager, the benchmark compares the observed holding to the cheapest comparable fund offered by the TDF manager itself. Our measure therefore captures the extent of price discrimination between TDF investors and non-TDF investors who purchase comparable underlying funds in the broader mutual fund market.

B.5 Measuring TDF wrapper fees

We define a TDF's wrapper fee as the difference between its total fee and the portfolio-weighted fee of its underlying funds, where the underlying-fund fee is computed using the benchmark described in Appendix B.4.

In our analysis, we interpret the wrapper fee as a measure of price discrimination between TDF investors and non-TDF investors who could purchase the relevant underlying funds directly. This interpretation is especially important after large fund families

Figure B.2: TDF wrapper fees: Baseline vs. Prospectus



Baseline wrapper fees vs. prospectus-based wrapper fees. Baseline wrapper fees are computed following the procedure described in Section 2.4. Prospectus wrapper fees are computed based on the net expense ratio of the underlying funds listed in the TDF prospectus.

began introducing internal zero-fee share classes. When those internal classes appear in the prospectus list of underlying funds, a wrapper-fee measure based directly on prospectus holdings can understate the effective cost of buying the same investment outside the TDF.

The importance of this restriction is visible in Figure B.2. Before 2017, our baseline measure exceeds the prospectus-based measure because it captures managers' choice of relatively expensive proprietary underlying funds. Beginning in 2017, however, several fund families introduced zero- or near-zero-fee internal share classes and the prospectus-based measure becomes larger because it treats those internal classes as if they were directly accessible to outside investors. Our benchmark instead continues to use accessible underlying funds, which yields an economically relevant measure of wrapper fees.

B.6 Merging with BrightScope plan data

We obtain data on 401(k) filings from 2009 to 2019 from BrightScope. In addition to plan menus and plan design features, BrightScope provides fund tickers for mutual funds. We use tickers to merge Morningstar fund-level information.

Because we lack information on characteristics of investment options classified as non-mutual funds(e.g., separate accounts, CITs etc.) we restrict our sample to 401(k) plans with more than 75% of assets allocated to mutual funds.

Figure C.1: Wrapper Fees vs Index Fund Fees

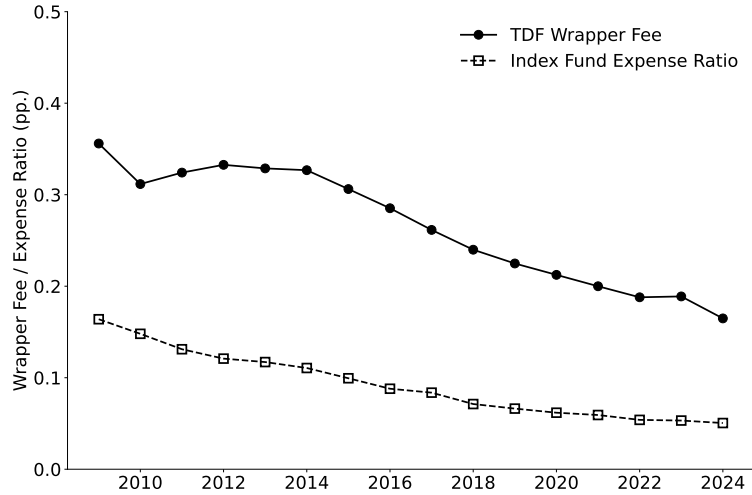


Figure plots the asset-weighted average wrapper fees and the asset-weighted average expense ratio of index funds. We restrict to index funds that invest in broad equity or fixed income securities, excluding those that invest in specific regions or sectors. Wrapper fees are constructed following the procedure described in Section 2.4.

C Cost and Value Added Explanations of Wrapper Fees

C.1 Index fund expense ratios as proxies for rebalancing management costs

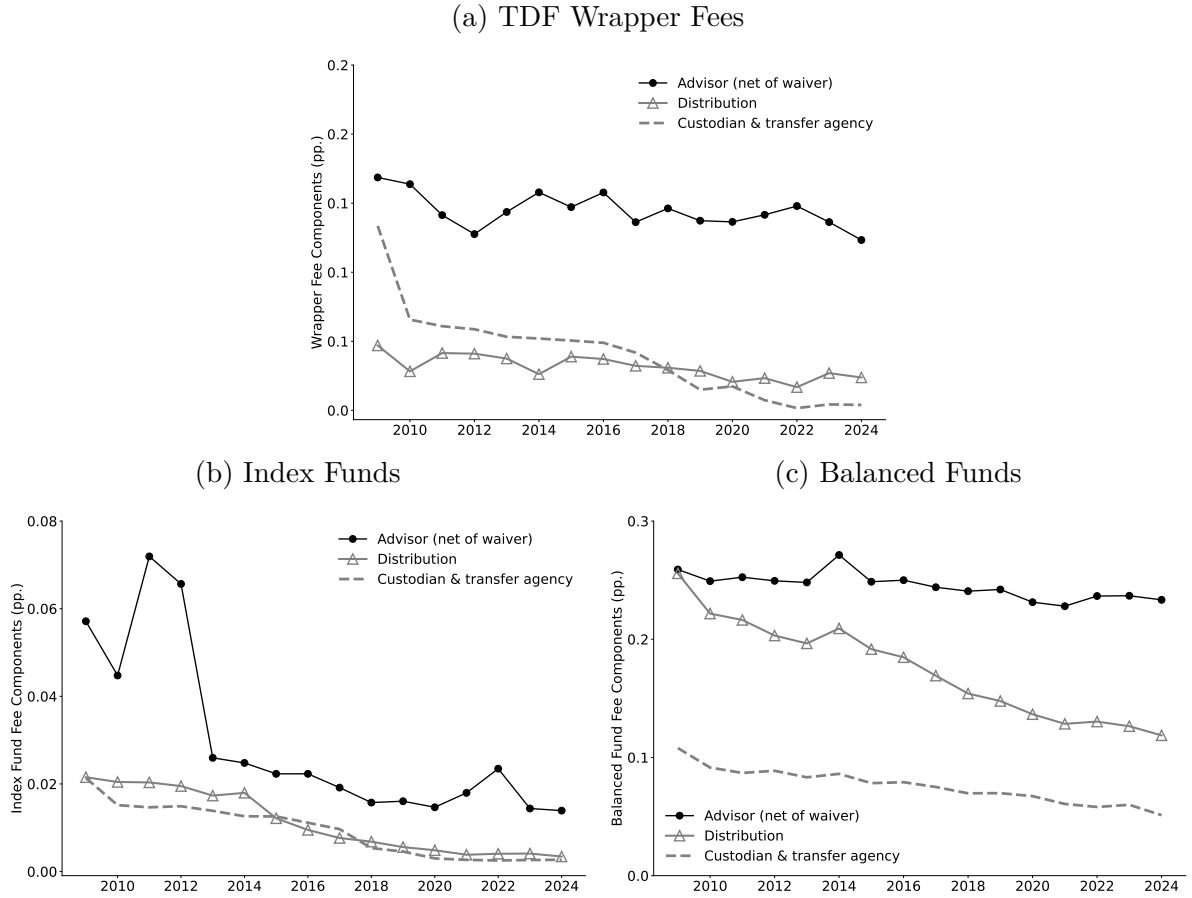
TDF managers rebalance underlying holdings of TDFs according to their glide paths, which can be costly. We highlight two different costs associated with rebalancing. The first consists of the costs TDF portfolio managers incur when managing the rebalancing operation, which we focus on here. The second is the transaction costs from rebalancing, which harm returns of TDFs but are not included in TDF expense ratios. We discuss this second cost in Appendix C.2.

We use index fund expense ratios to proxy for such rebalancing management costs. Index fund portfolio managers rebalance underlying constituents of their funds according to their benchmark, which is similar to how TDF managers rebalance following their glide paths. The operation of index fund managers may require more effort, since they rebalance over a larger number of individual stocks or bonds rather than 5 to 20 underlying funds held by TDFs (Table 1). In addition, index fund expense ratios include other costs and markups, and we consider them to be upper bounds of rebalancing management costs.

Figure C.1 shows that TDF wrapper fees are consistently higher than index fund expense ratios. The asset-weighted index fund expense ratio is around 7 bps as of 2019²⁷,

²⁷We include broad investment categories on US or global equities and fixed income, excluding index

Figure C.2: Decomposition of TDF Wrapper Fees and Index Fund Expense Ratios



Breakdown of fee components: advisor fees, distribution fees, and custodian and transfer agency fees. Panel (a) shows asset-weighted average of each fee component for TDF wrapper fees. The construction of these components follows the construction of our baseline wrapper fee. Panels (b) and (c) show breakdown for index funds and balanced funds. We restrict to the same set of fund managers that have TDFs.

whereas TDF wrapper fees are three times as large. This comparison suggests that costs of managing life-cycle rebalancing are likely much smaller than implied by wrapper fees.

In Figure C.2, we also compare different components of wrapper fees. Figure C.2a shows that the advisor fee, which compensates for portfolio management, is the largest component of wrapper fees. Advisor fees remain roughly constant around 10 bps in our sample. This pattern is difficult to reconcile with a cost-based explanation. Because life-cycle rebalancing follows a predetermined glide path, its costs should be relatively fixed in dollar terms and become smaller fractions of fund assets. To the contrary, since advisor fees are paid to internal portfolio managers, fund managers can have more discretion in using this component to charge markups. As a comparison, Figure C.2b shows that advisor fees for index funds have declined over time and are substantially lower than the advisor-fee component of TDFs.

funds that focus on specific countries or sectors.

C.2 Transaction costs of life-cycle rebalancing

The rebalancing of selling equities and buying bonds generates transaction costs ([Harvey et al., 2025](#)). Such costs affect fund performance and are not reflected in fund expense ratios. For example, the prospectus of Vanguard Target Retirement fund clearly states: *“The Fund pays transaction costs, such as purchase fees, when it buys and sells securities... These costs, which are not reflected in annual fund operating expenses or in the example, affect the Fund’s performance.”*

Relatedly, one potential value added of TDFs is that managers optimize rebalancing to minimize transaction costs, thereby increasing TDF returns. For example, Vanguard follows a dynamic rebalancing strategy based on asset allocation thresholds ([Vanguard, 2024](#)). In simulation exercises, the threshold-based rebalancing reduces transaction costs by 13.2 bps and increases relative return by 11 bps, relative to quarterly rebalancing. Presumably, other large asset management firms that manage TDFs also optimize their rebalancing, and so the transaction cost reduction should be competed away. It is unlikely that TDF managers are able to pass on a large fraction of this transaction cost reduction to investors.

It is worth noting that TDF managers do not engage in fund research and selection, compared to those who manage funds of hedge funds for example. While funds of funds generally charge an additional layer of fees, such additional layer of fees for TDFs should only capture life-cycle rebalancing. Previous pieces of evidence suggest that managing life-cycle rebalancing should involve low costs and generate limited value added relative to the magnitude of wrapper fees.

C.3 Wrapper fees of index vs. active TDFs

We can also shed light on cost explanations by comparing wrapper fees between active and index TDFs. Active and index TDFs differ in whether they invest in active or index underlying funds, but managing life-cycle rebalancing over active and index funds should be similar (especially in tax-exempt retirement accounts without capital gains tax considerations). However, [Figure C.3](#) shows that index TDFs have lower wrapper fees compared to active TDFs. In regressions including fund manager fixed effects and other controls, [Appendix Table D.1](#) shows that index TDFs have on average 10 bps lower wrapper fees than active TDFs. As a comparison, other characteristics of underlying funds are either statistically insignificant or have small economic magnitudes. This difference in wrapper fees is thus unlikely to reflect costs of managing life-cycle rebalancing, but instead captures the relatively higher fee-sensitivity of index fund investors.

Figure C.3: Wrapper Fees: Active vs. Index

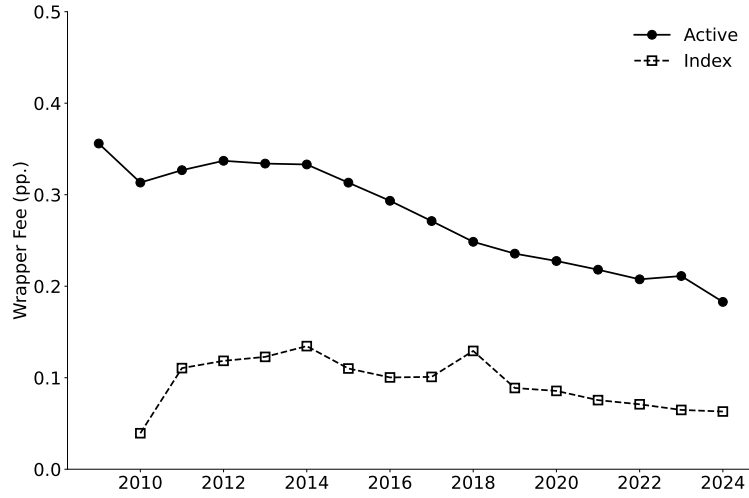


Figure plots TDF wrapper fees separately for active and index TDFs. Wrapper fees are constructed following the procedure described in Section 2.4.

C.4 TDF wrapper fees in Thrift Savings Plan

We also look at expense ratios of TDFs and other funds in the Thrift Savings Plan (TSP) for federal employees. TSP funds are overseen by the Federal Retirement Thrift Investment Board and managed by BlackRock and State Street. BlackRock and State Street offer core index funds in different asset classes, as well as TDFs that mix these core funds based on retirement years. These funds are only offered to investors in TSPs.

Table C.1 shows the funds offered and their expense ratios. All funds are index funds. TSP is the largest defined contribution plan. Thanks to its scale, fund expense ratios are generally low. TDF fees range from 3.5 bps for investors already retired to 4.1 bps for investors expected to retire around 2070. As a comparison, the expense ratios of individual funds range from 3.4 to 5.1 bps. TDF fees are close to weighted averages of fees of other funds, where funds with target retirement year of 2070 are slightly more expensive due to higher allocation to small cap and international stock funds.

Hence, it appears feasible for fund managers to charge zero wrapper fees, which is inconsistent with costs of managing TDFs.

C.5 Recordkeeping costs

Another explanation of the wrapper fee is that they partially compensate for the costs of providing 401(k) services, especially for integrated managers who both offer mutual funds to 401(k) investors and provide recordkeeping services. The total fees these integrated managers receive from 401(k) plans include both expense ratios of mutual funds on those plan menus and also direct recordkeeping fees. Hence, higher wrapper fees from TDFs could potentially allow these integrated managers to reduce direct recordkeeping fees

Table C.1: TSP Fund Expense Ratios

Fund	Expense Ratio (pp.)
TDFs	
Income	0.035
2030	0.039
2040	0.040
2050	0.041
2060	0.041
2070	0.041
Government securities	0.034
US fixed income	0.035
US large cap	0.035
US small cap	0.051
International stock	0.048

Expense ratios of funds offered in Thrift Savings Plan (TSP) as of 2026. <https://www.tsp.gov/funds-individual/>

Table C.2: TDF Wrapper Fees to Offset Recordkeeping Fees

	Direct Recordkeeping Fees (pp.)				
Proprietary TDF	-0.02*** (0.00)	-0.03*** (0.00)		-0.02*** (0.00)	
Frac Proprietary Funds		-0.02*** (0.00)	-0.01*** (0.00)	-0.04*** (0.00)	-0.02*** (0.00)
Proprietary TDF $\times$ Wrapper (pp.)			-0.09*** (0.00)		-0.08*** (0.00)
Any TDF	-0.01*** (0.00)	0.00** (0.00)	0.00*** (0.00)		
log(Plan assets)		-0.09*** (0.00)	-0.09*** (0.00)	-0.08*** (0.00)	-0.08*** (0.00)
log(Plan participants)		0.07*** (0.00)	0.07*** (0.00)	0.06*** (0.00)	0.06*** (0.00)
N	423,866	423,866	423,866	220,679	220,679
R2	0.20	0.28	0.28	0.34	0.34
Recordkeeper $\times$ Year FE	X	X	X	X	X
Plan Size Group FE		X	X	X	X

Regression of direct recordkeeping fees (measured as percentage points over plan assets) on whether the recordkeeper provides its own TDF on plan menus. The first three columns include all observations. The final two columns restrict to plans that offer TDFs. Our sample ranges from 2009 to 2019.

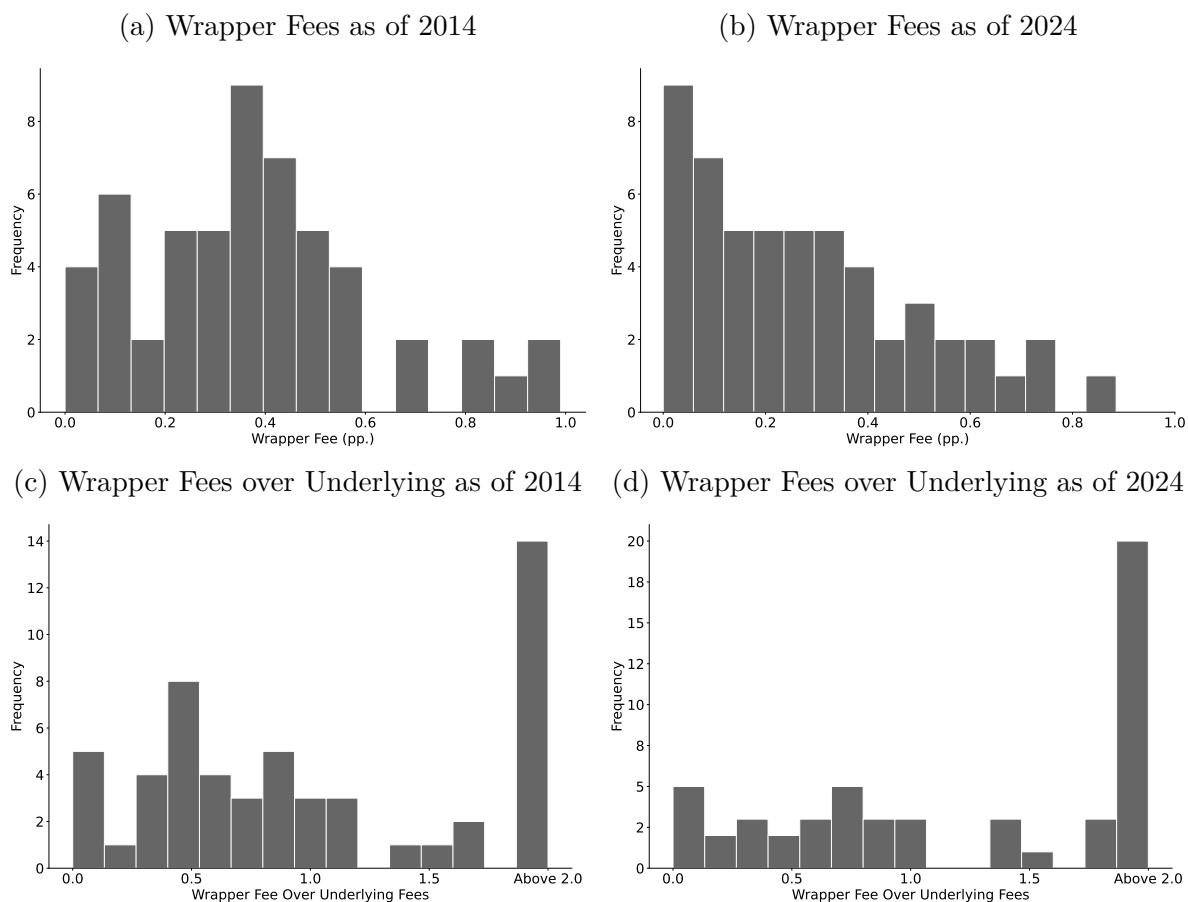
while keeping total plan fees unchanged. While expense ratios of non-TDFs can cover recordkeeping fees, using TDFs can be more effective since a higher fraction of TDF assets comes from 401(k) plans.

To assess this explanation, we compare direct recordkeeping fees with and without proprietary TDFs. We compute direct recordkeeping fees as a percentage of plan assets to have the same units as fund fees. We include an indicator that is equal to zero if the recordkeeper is not integrated without any proprietary mutual funds, or if the recordkeeper is integrated but the TDF in this plan is managed by another firm. Table C.2 shows that direct recordkeeping fees are 2 to 3 bps lower when recordkeepers offer their proprietary TDFs. From the second column, we control for the fraction of any proprietary funds, so we examine the effect of TDF on top of how integrated managers use other funds to cover recordkeeping fees. In general, the compensation for recordkeeping costs is small relative to the magnitude of TDF wrapper fees. In the third and fifth columns, we replace the indicator of proprietary TDF with its interaction with TDF wrapper fees to focus on the intensive margin. The estimates suggest that a 1 pp increase in wrapper fee is associated with less than 0.1 pp decrease in direct recordkeeping fees. In other words, potential compensation for recordkeeping fees accounts for less than 10% of TDF wrapper fees.

We note that this regression shows the correlation between proprietary TDF and direct recordkeeping fees. There are potentially other explanations of this coefficient. For example, there can be a selection effect where employers with proprietary TDFs are different, which leads to lower recordkeeping fees. Our plan size control may partially but not fully address such selection. To the extent these regression results are consistent with recordkeeping cost offset, it suggests such cost offset is small in magnitude compared to wrapper fees.

D Additional Figures and Tables

Figure D.1: TDF Wrapper Fees Distribution



Top panel plots the distribution of wrapper fees for the year 2014 on the left and for the year 2024 on the right. Bottom panel plots the distribution of wrapper-to-underlying fee ratio for the year 2014 (left) and for the year 2024 (right).

Figure D.2: TDF Total Fees and Underlying Fees

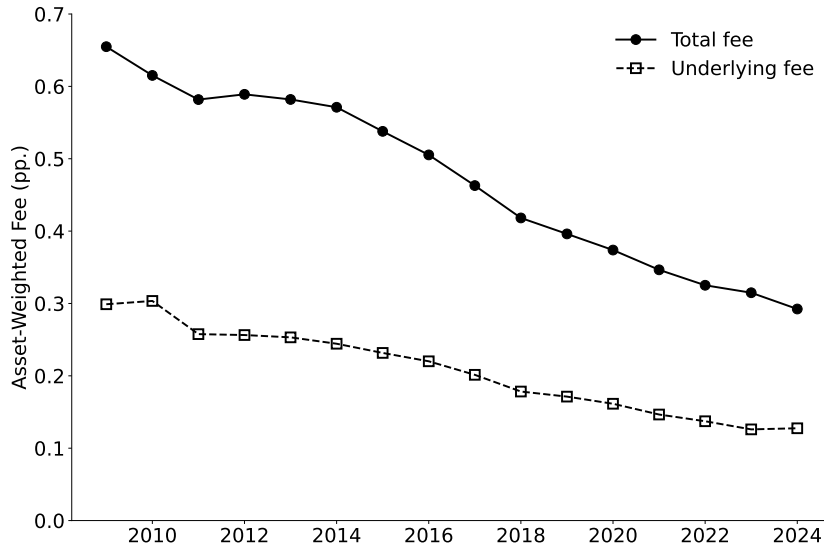
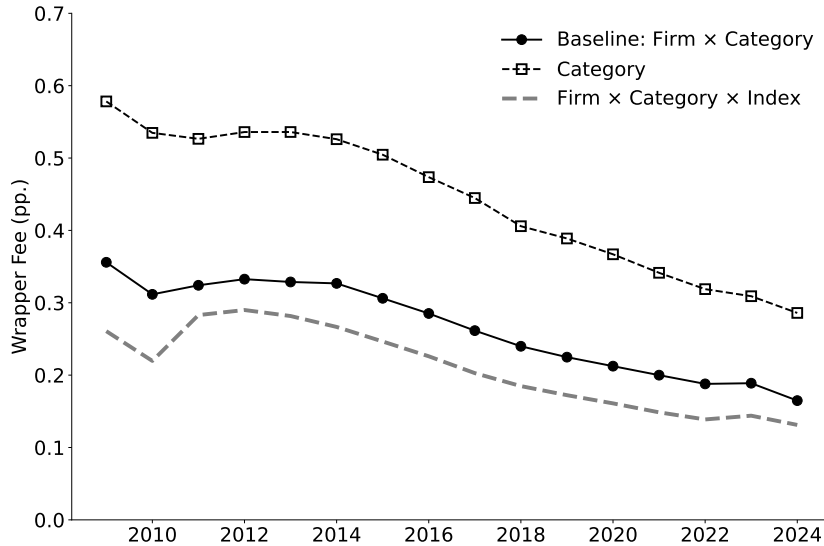


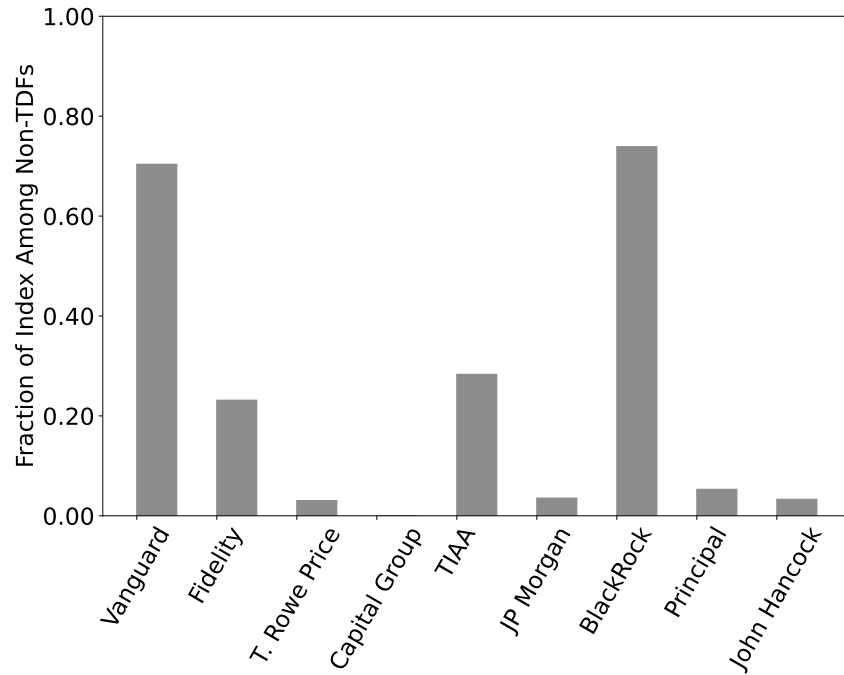
Figure plots the asset-weighted average expense ratios of Target-Date Funds (TDFs) and underlying funds held by TDFs. Underlying funds are identified based on the procedure described in Section 2.4.

Figure D.3: Alternative Wrapper Fees



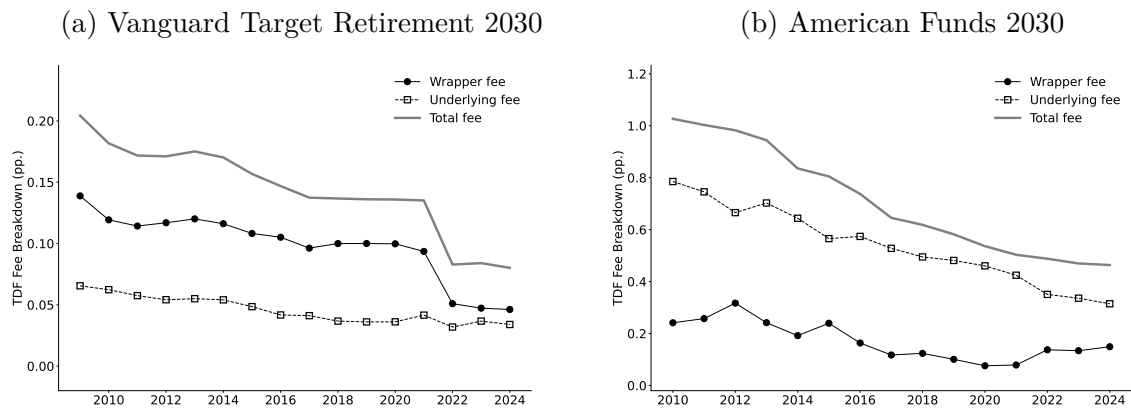
Asset-weighted average wrapper fees of Target-Date Funds (TDFs). The baseline definition replaces the prospectus underlying funds with the cheapest accessible fund within the same firm $\times$ Morningstar global category. The alternative definition applies the same procedure within the same Morningstar global category, and firm $\times$ Morningstar global category $\times$ index/active cohort.

Figure D.4: Index Funds Among Non-TDFs



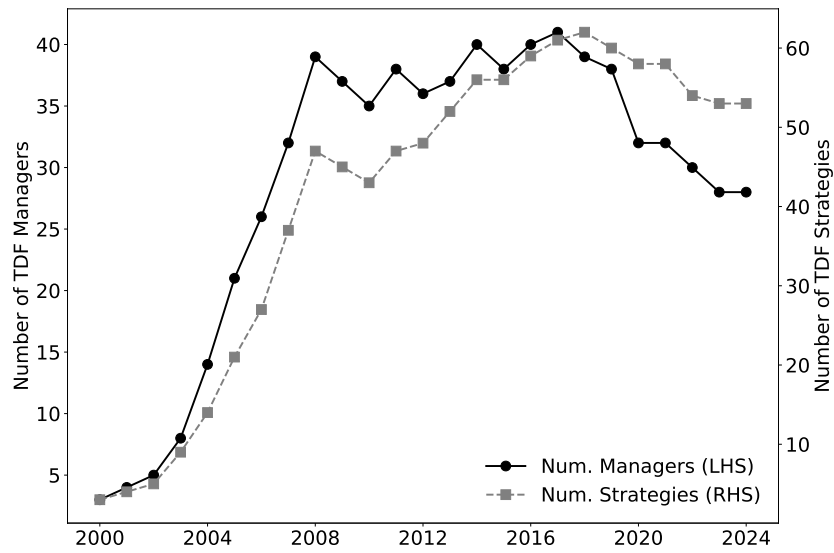
Fraction of index funds among non-TDFs of largest TDF managers as of 2019.

Figure D.5: Additional Wrapper Fee Examples



Panel (a) shows TDF fees, wrapper fees and underlying fees for Vanguard Target Retirement 2030. Panel (b) shows TDF fees, wrapper fees and underlying fees for American Funds 2030.

Figure D.6: Number of TDF Managers and Strategies



Number of TDF fund managers and TDF strategies each year.

Table D.1: Variation of TDF Wrapper Fees

	Wrapper Fee (pp.)			
Log(AUM)	-0.006*	0.001	-0.001	0.013***
	(0.003)	(0.004)	(0.004)	(0.005)
Index Fund	-0.122***	-0.121***	-0.130***	-0.111***
	(0.013)	(0.017)	(0.018)	(0.026)
Log(Number of Underlying Funds)		-0.000	-0.000	0.025*
		(0.013)	(0.011)	(0.014)
Proprietary Underlying Funds (%)		-0.001***	-0.001***	-0.001
		(0.000)	(0.000)	(0.000)
Recordkeeper		-0.044***	-0.050***	
		(0.015)	(0.018)	
Equity Share at Retirement (%)			0.002**	0.004***
			(0.001)	(0.001)
Initial Equity Share (%)			0.000	-0.001*
			(0.000)	(0.001)
N	812	812	738	736
R2	0.06	0.10	0.11	0.60
Year FE	X	X	X	X
Firm FE				X

Fund $\times$ year level OLS regressions of TDF wrapper fees on fund characteristics.

Table D.2: Employer Demand Estimates

	Employer Default Choice							
	<200		200-499		500-999		>1000	
	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV
Expense Ratio (pp)	-3.30***	-0.93***	-3.73***	-1.76***	-4.10***	-1.73***	-5.17***	-3.38***
	(0.05)	(0.05)	(0.06)	(0.05)	(0.09)	(0.08)	(0.09)	(0.07)
Recordkeepers	1.57***	1.00***	1.39***	0.71***	1.26***	0.42***	1.12***	-0.34***
	(0.01)	(0.02)	(0.01)	(0.02)	(0.02)	(0.03)	(0.02)	(0.03)
log(Fund Age)	1.81***	1.92***	2.12***	2.15***	2.48***	2.44***	3.30***	3.34***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.03)	(0.03)	(0.03)	(0.03)
Past Returns (pp)	0.18***	0.15***	0.19***	0.16***	0.21***	0.15***	0.21***	0.13***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Final Equity		-0.05***		-0.03***		-0.02***		0.01***
		(0.00)		(0.00)		(0.00)		(0.00)
Max Equity		0.04***		0.03***		0.03***		0.02***
		(0.00)		(0.00)		(0.00)		(0.00)
N	2,071,364	2,071,364	1,951,474	1,951,474	769,609	769,609	972,400	972,400
Elasticity	-1.16	-0.33	-1.34	-0.64	-1.48	-0.62	-1.76	-1.15
Year FE	X	X	X	X	X	X	X	X

MLE estimates of Equation (2). We restrict to plans with over 75% of assets invested in mutual funds. We also drop plans whose TDF is not a mutual fund. The cost-shifter instrument is based on custodian and transfer agency costs. The differentiation IV is based on differences in TDF glide paths captured by highest equity allocations and terminal equity allocations.

Table D.3: Demand Estimates for 401(k) TDF Investors

	401(k) Investor Allocation to Default							
	<200		200-499		500-999		>1000	
	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV
Expense Ratio (pp)	-0.26** (0.12)	-0.67*** (0.08)	-0.80*** (0.12)	-0.78*** (0.07)	-0.76*** (0.17)	-0.83*** (0.10)	-0.55*** (0.12)	-0.45*** (0.08)
log(Fund Age)	0.64*** (0.03)	0.81*** (0.03)	0.83*** (0.03)	0.91*** (0.03)	0.76*** (0.05)	0.80*** (0.04)	0.45*** (0.06)	0.43*** (0.04)
Default Option	0.43*** (0.02)	0.42*** (0.02)	0.51*** (0.02)	0.50*** (0.02)	0.41*** (0.03)	0.40*** (0.03)	0.29*** (0.02)	0.29*** (0.02)
Auto-Enroll	0.34*** (0.02)	0.34*** (0.02)	0.38*** (0.01)	0.38*** (0.01)	0.42*** (0.02)	0.42*** (0.02)	0.37*** (0.02)	0.36*** (0.02)
Recordkeepers	0.19*** (0.03)	0.01 (0.03)	-0.05 (0.04)	-0.10*** (0.03)	-0.06 (0.06)	-0.11*** (0.04)	-0.10** (0.05)	-0.12*** (0.04)
Past Returns (pp)	-0.02*** (0.01)	0.00 (0.01)	0.01** (0.01)	0.03*** (0.00)	0.02*** (0.01)	0.03*** (0.01)	0.03*** (0.01)	0.02*** (0.01)
log(Plan Partcp)	0.45*** (0.03)	0.45*** (0.03)	0.41*** (0.02)	0.41*** (0.02)	0.45*** (0.04)	0.45*** (0.04)	0.32*** (0.01)	0.32*** (0.01)
log(Plan Assets)	-0.40*** (0.01)	-0.41*** (0.01)	-0.42*** (0.01)	-0.41*** (0.01)	-0.41*** (0.01)	-0.41*** (0.01)	-0.39*** (0.01)	-0.39*** (0.01)
N	70,403	70,403	72,803	72,803	32,864	32,864	42,758	42,758
Elasticity	-0.07	-0.17	-0.21	-0.20	-0.20	-0.22	-0.15	-0.12
Year FE	X	X	X	X	X	X	X	X

Estimates of Equation (3) at the plan-year level. We restrict the sample to plans that offer a mutual fund TDF series. The cost-shifter instrument is based on custodian and transfer agency costs. The differentiation IV is based on differences in TDF glide paths captured by highest equity allocations and terminal equity allocations.

Table D.4: Demand Estimates for Non-TDF Investors

	Inside 401(k)								Outside 401(k)	
	<200		200-499		500-999		>1000		Outside	
	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV	Cost IV	Diff IV
Expense Ratio (pp)	-1.66*** (0.16)	-5.09*** (1.28)	-2.10*** (0.18)	-5.56*** (1.09)	-2.80*** (0.20)	-5.34*** (1.31)	-3.97*** (0.20)	-7.61*** (1.26)	-6.87*** (0.18)	-2.24*** (0.62)
log(Fund Age)	1.61*** (0.02)	1.67*** (0.03)	1.68*** (0.03)	1.73*** (0.03)	1.56*** (0.03)	1.61*** (0.04)	1.63*** (0.03)	1.66*** (0.04)	1.50*** (0.02)	1.36*** (0.02)
Past Returns (pp)	0.02*** (0.00)	0.02*** (0.00)	0.01*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)
N	29,824	29,824	29,498	29,498	24,751	24,751	27,448	27,448	56,286	56,286
Elasticity	-1.06	-3.26	-1.31	-3.48	-1.68	-3.21	-2.09	-4.00	-3.98	-1.30
Category $\times$ Year $\times$ Index FE	X	X	X	X	X	X	X	X	X	X
Fund manager FE	X	X	X	X	X	X	X	X	X	X

Estimates of Equation (4) at the fund-year level with year $\times$ Morningstar category $\times$ index fixed effects. In columns (1) to (8) under “Inside 401(k)”, we estimate demand across funds in 401(k) plans. In the last two columns under “Outside 401(k)”, we estimate demand across funds outside 401(k) plans. The cost-shifter instrument is based on custodian and transfer agency costs. The differentiation IV is based on differences in past returns.

Table D.5: Demand Estimates for 401(k) TDF Investors, Control for Non-TDF Options

	401(k) Investor Allocation to Default							
	<200		200-499		500-999		>1000	
	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV	OLS	Hausman IV
Expense Ratio (pp)	0.04 (0.03)	-0.20*** (0.03)	-0.04 (0.02)	-0.28*** (0.03)	-0.17*** (0.03)	-0.41*** (0.04)	-0.06* (0.03)	-0.20*** (0.03)
log(Fund Age)	0.65*** (0.03)	0.66*** (0.03)	0.73*** (0.03)	0.76*** (0.03)	0.63*** (0.04)	0.69*** (0.04)	0.27*** (0.04)	0.32*** (0.04)
Default Option	0.43*** (0.02)	0.43*** (0.02)	0.48*** (0.02)	0.48*** (0.02)	0.39*** (0.03)	0.39*** (0.03)	0.28*** (0.02)	0.28*** (0.02)
Auto-Enroll	0.34*** (0.02)	0.34*** (0.02)	0.39*** (0.01)	0.38*** (0.01)	0.42*** (0.02)	0.42*** (0.02)	0.36*** (0.02)	0.36*** (0.02)
Recordkeepers	0.23*** (0.02)	0.18*** (0.02)	0.13*** (0.02)	0.07*** (0.02)	0.10*** (0.03)	0.03 (0.03)	0.08*** (0.03)	0.03 (0.03)
Past Returns (pp)	-0.03*** (0.00)	-0.02*** (0.00)	-0.01** (0.00)	-0.00 (0.00)	0.01 (0.01)	0.01** (0.01)	0.01*** (0.01)	0.02*** (0.01)
log(Plan Partcp)	0.46*** (0.03)	0.46*** (0.03)	0.40*** (0.02)	0.40*** (0.02)	0.44*** (0.04)	0.45*** (0.04)	0.31*** (0.01)	0.31*** (0.01)
log(Plan Assets)	-0.38*** (0.01)	-0.39*** (0.01)	-0.38*** (0.01)	-0.39*** (0.01)	-0.38*** (0.01)	-0.39*** (0.01)	-0.37*** (0.01)	-0.38*** (0.01)
Inclusive Value Non-TDF	-0.01 (0.01)	-0.01 (0.01)	0.04*** (0.01)	0.04*** (0.01)	0.05*** (0.01)	0.05*** (0.01)	0.04*** (0.01)	0.04*** (0.01)
N	67,365	67,365	70,584	70,584	32,228	32,228	42,109	42,109
Elasticity	0.01	-0.05	-0.01	-0.07	-0.05	-0.11	-0.02	-0.05
Year FE	X	X	X	X	X	X	X	X

Estimates of Equation (3) at the plan-year level. We restrict the sample to plans that offer a mutual fund TDF series. We also control for inclusive value of non-TDF options on the same plan menus. Inclusive value is defined as $\ln\left(1 + \sum_j \exp(\delta_j)\right) = \ln\left(1 + \sum_j \exp(\ln(s_j) - \ln(s_0))\right)$ where s_j denotes the share of plan assets allocated to non-TDF j and s_0 the share allocated to any cash funds. We normalize the utility of cash funds to zero.